



US012410613B2

(12) **United States Patent**  
**Collins et al.**

(10) **Patent No.:** **US 12,410,613 B2**  
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **PRE-MANUFACTURED LOAD BEARING WALLS FOR A MULTI-STORY BUILDING**

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(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 186 days.

(21) Appl. No.: **18/248,961**

(22) PCT Filed: **Oct. 21, 2021**

(86) PCT No.: **PCT/US2021/056077**

§ 371 (c)(1),  
(2) Date: **Apr. 13, 2023**

(87) PCT Pub. No.: **WO2022/087283**

PCT Pub. Date: **Apr. 28, 2022**

(65) **Prior Publication Data**

US 2023/0417049 A1 Dec. 28, 2023

**Related U.S. Application Data**

(60) Provisional application No. 63/178,515, filed on Apr. 22, 2021, provisional application No. 63/104,239, filed on Oct. 22, 2020.

(51) **Int. Cl.**  
**E04B 1/08** (2006.01)  
**B66B 11/00** (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ..... **E04B 5/04** (2013.01); **B66B 11/0005**  
(2013.01); **E04B 1/02** (2013.01); **E04B 1/08**  
(2013.01);

(Continued)

(58) **Field of Classification Search**

CPC ..... **E04B 2001/2481**; **E04B 9/008**; **E04B 1/2403**; **E04B 1/24**; **E04B 1/18**  
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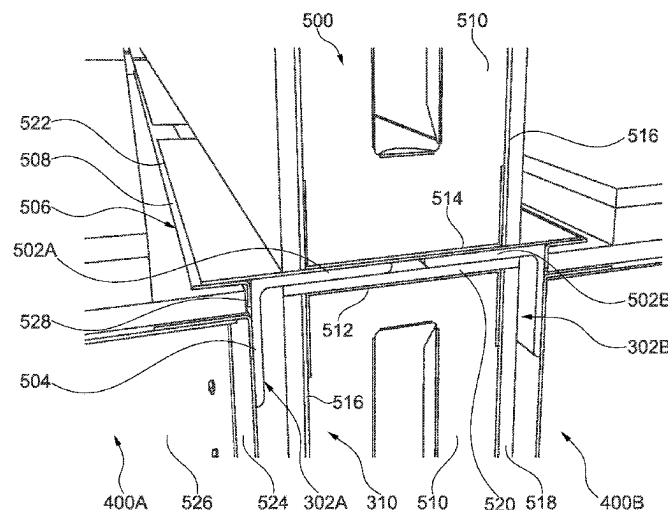
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(57) **ABSTRACT**

A multi-story building is provided with load bearing walls that are able to withstand vertical loads and lateral loads. The building may be a low-rise building or a mid-rise building. The load bearing walls, as well as floor-ceiling panels, corridor panels, utility walls, and other parts of the building, are pre-manufactured off-site and then installed on-site at the site of the building. The floor-ceiling panels are hung from the load bearing walls and are capable to receive and transfer lateral and vertical loads, as well as providing improved sound proofing and fire rating for the building.

**20 Claims, 15 Drawing Sheets**



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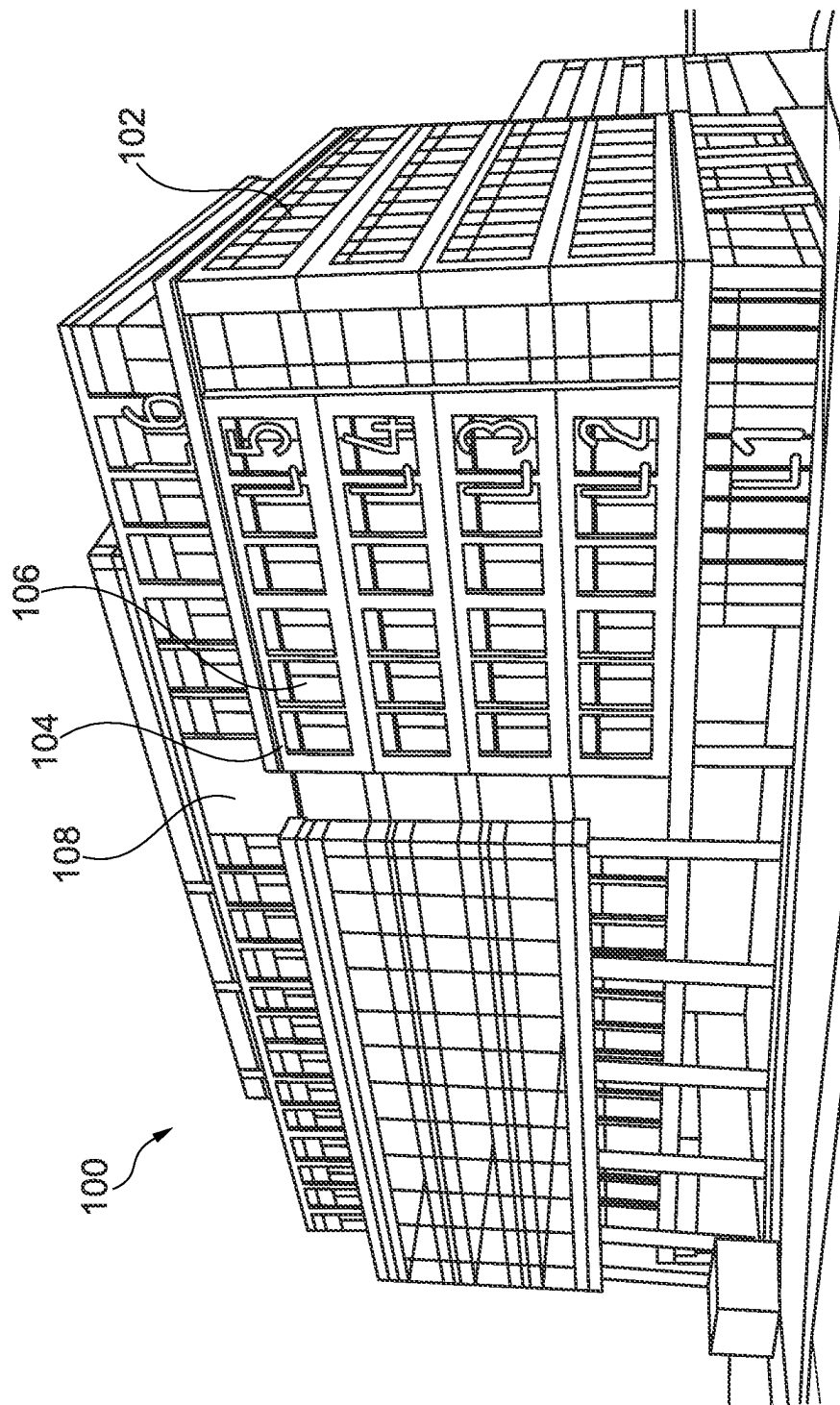
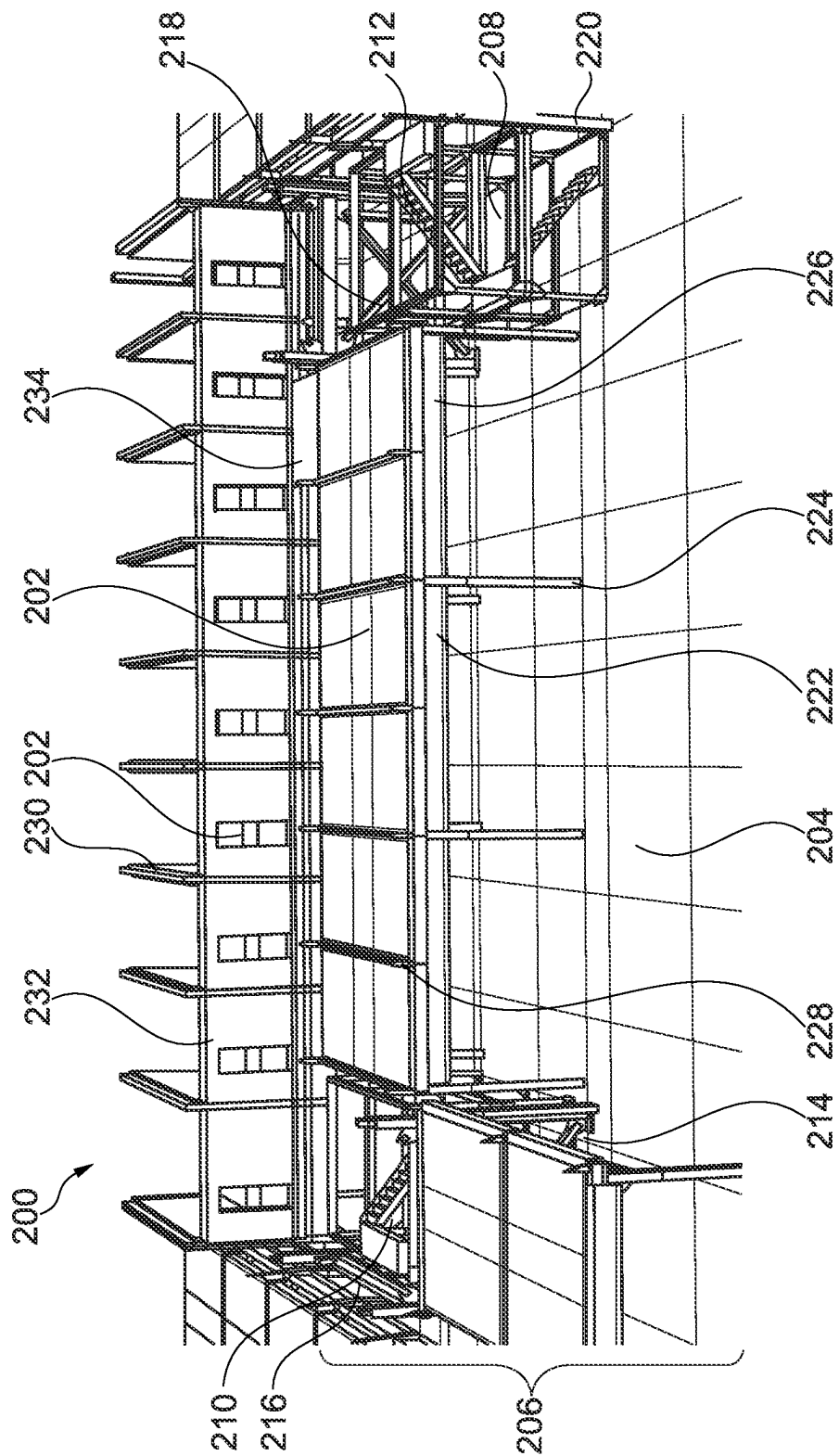
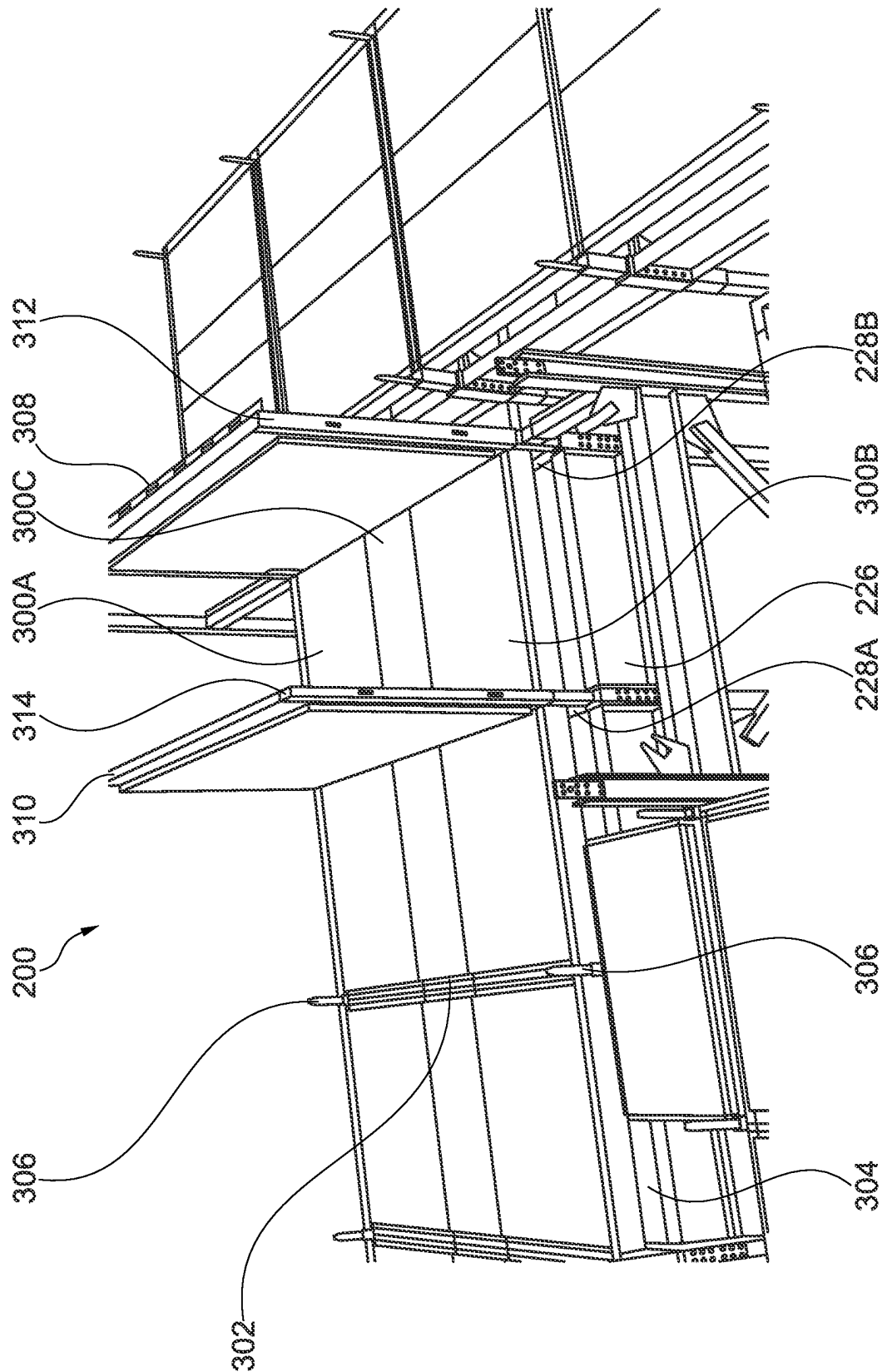


FIG. 1



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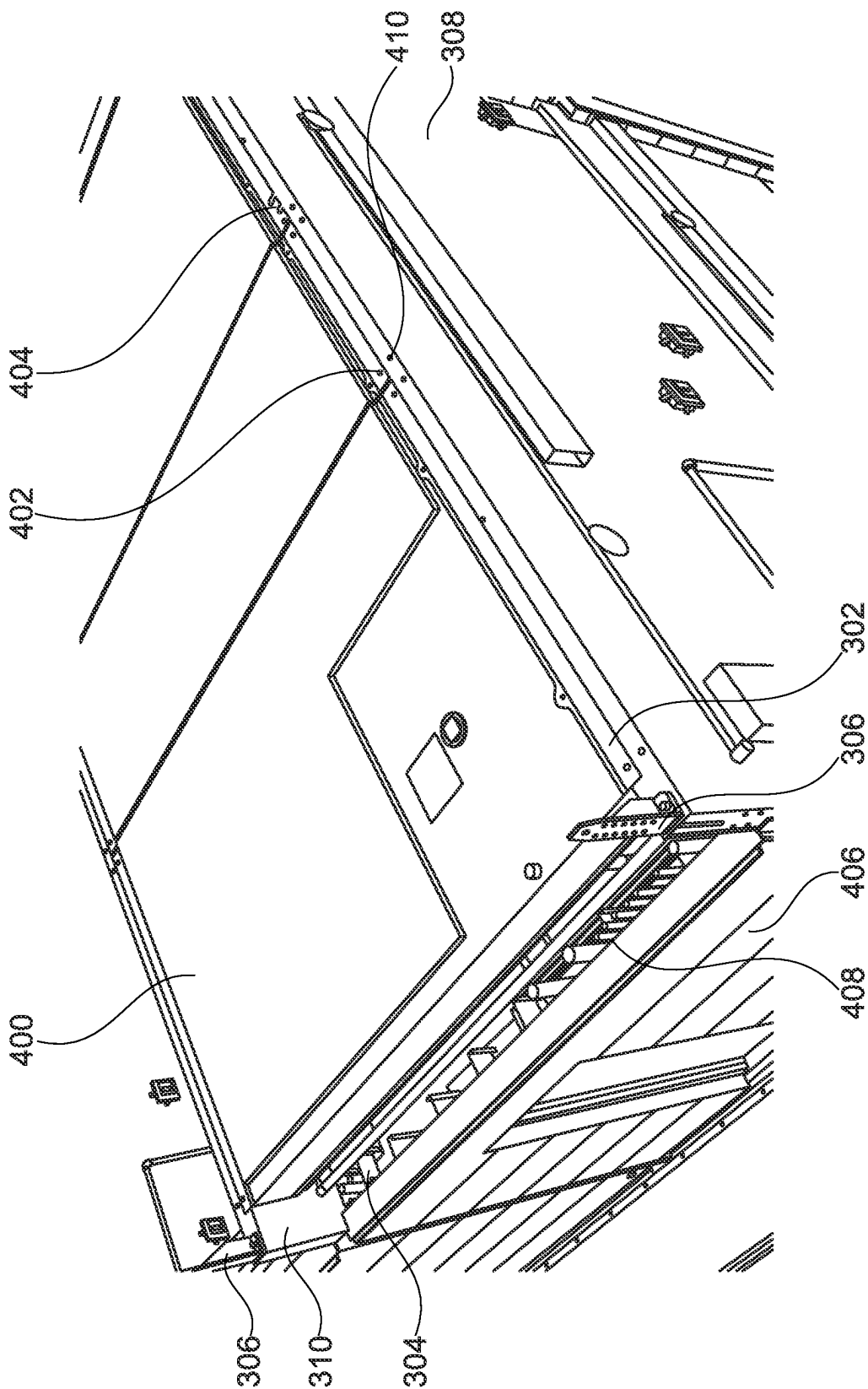
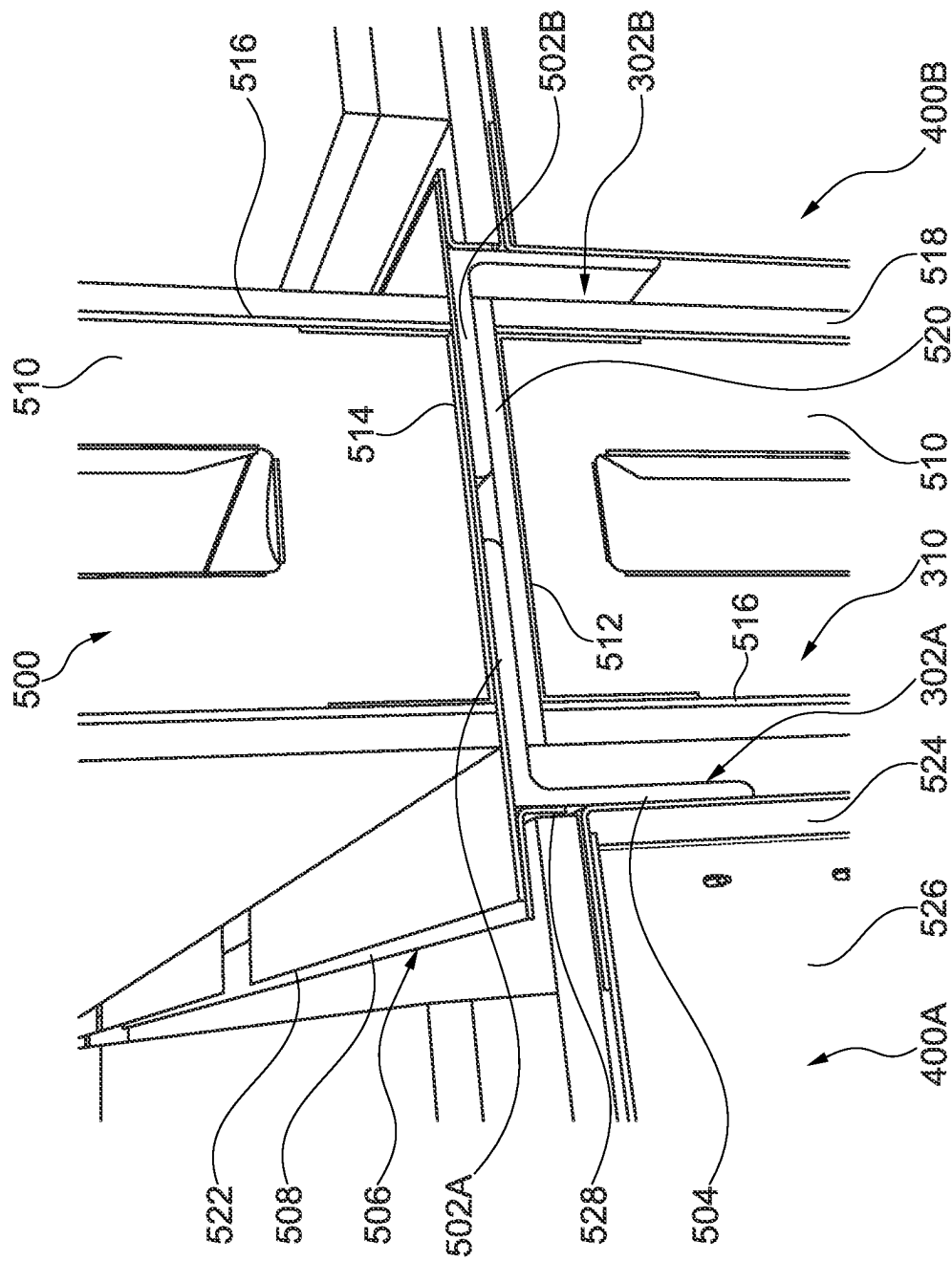


FIG. 4



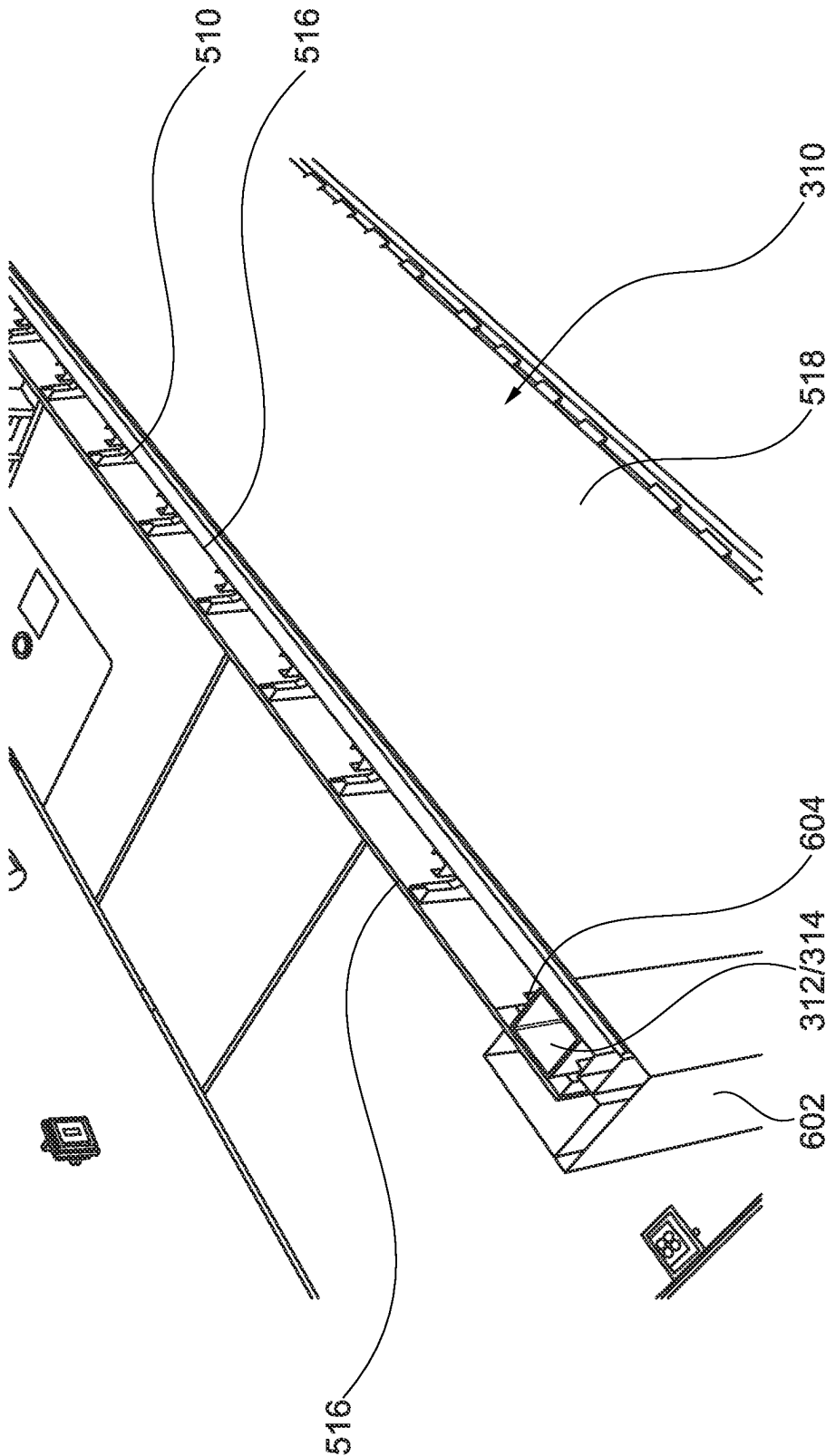


FIG. 6



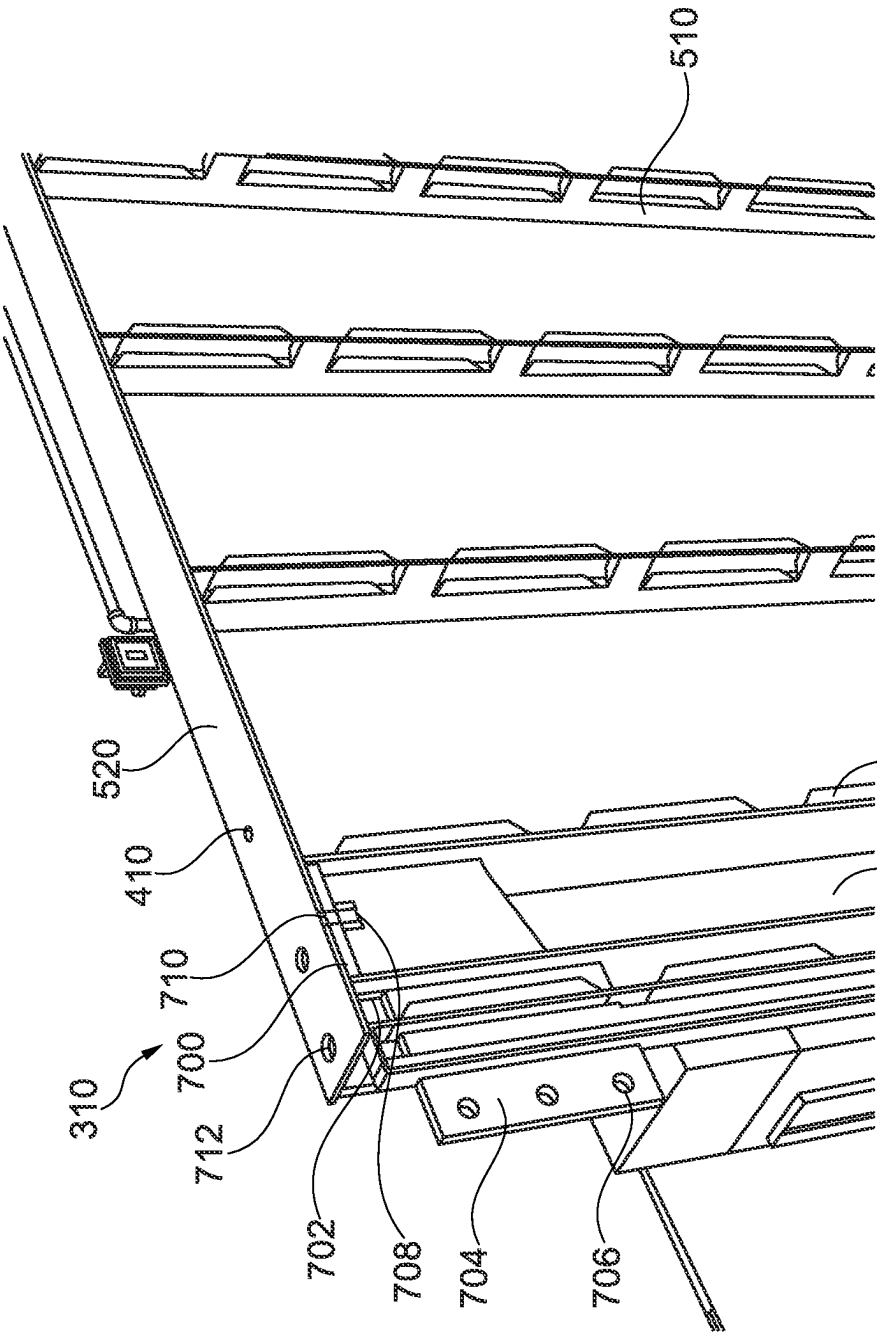


FIG. 7

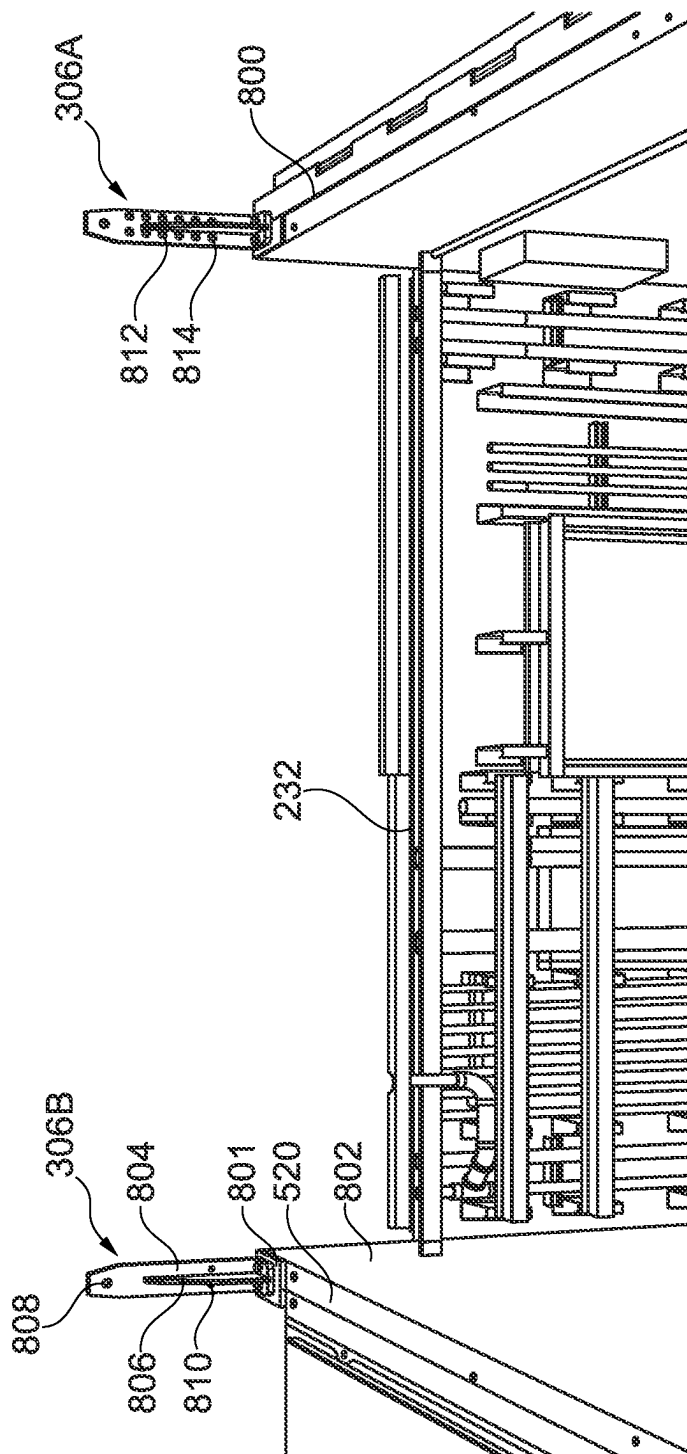


FIG. 8

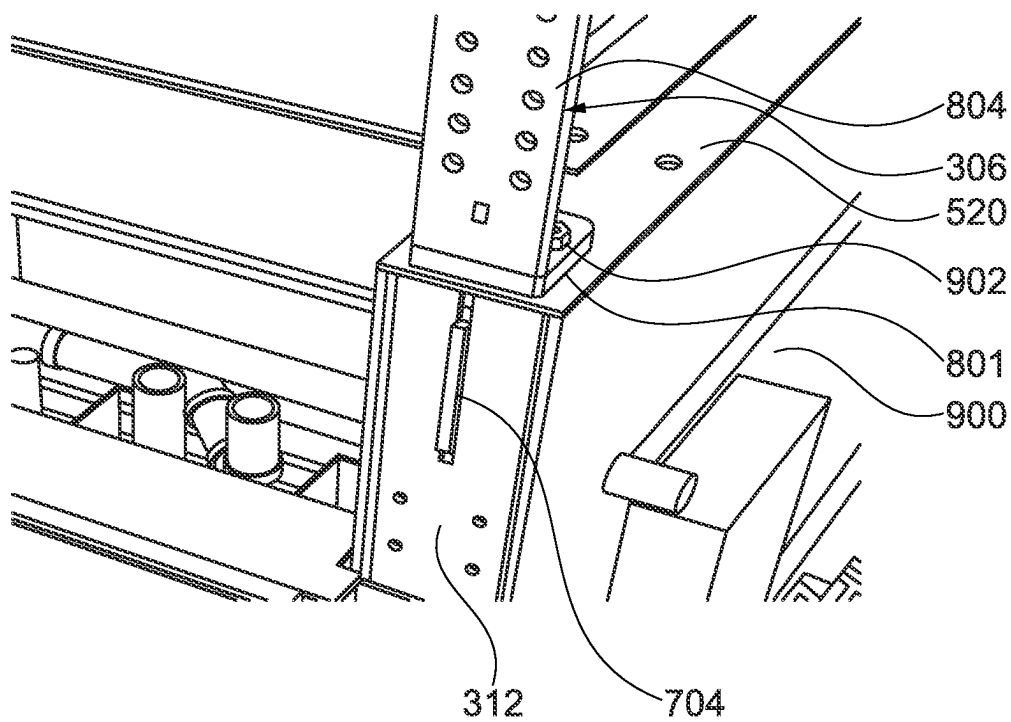


FIG. 9

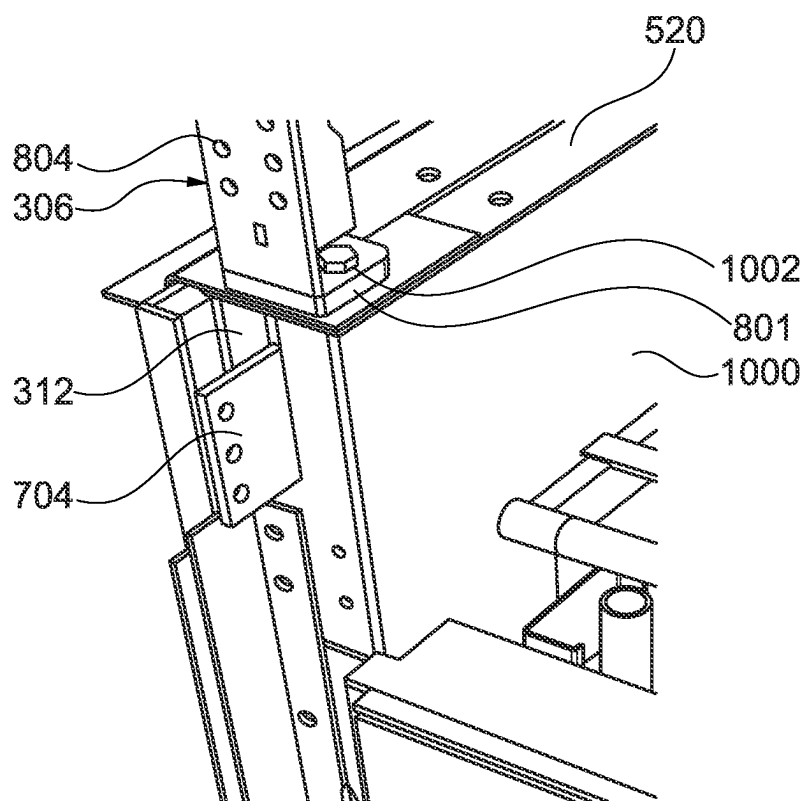


FIG. 10

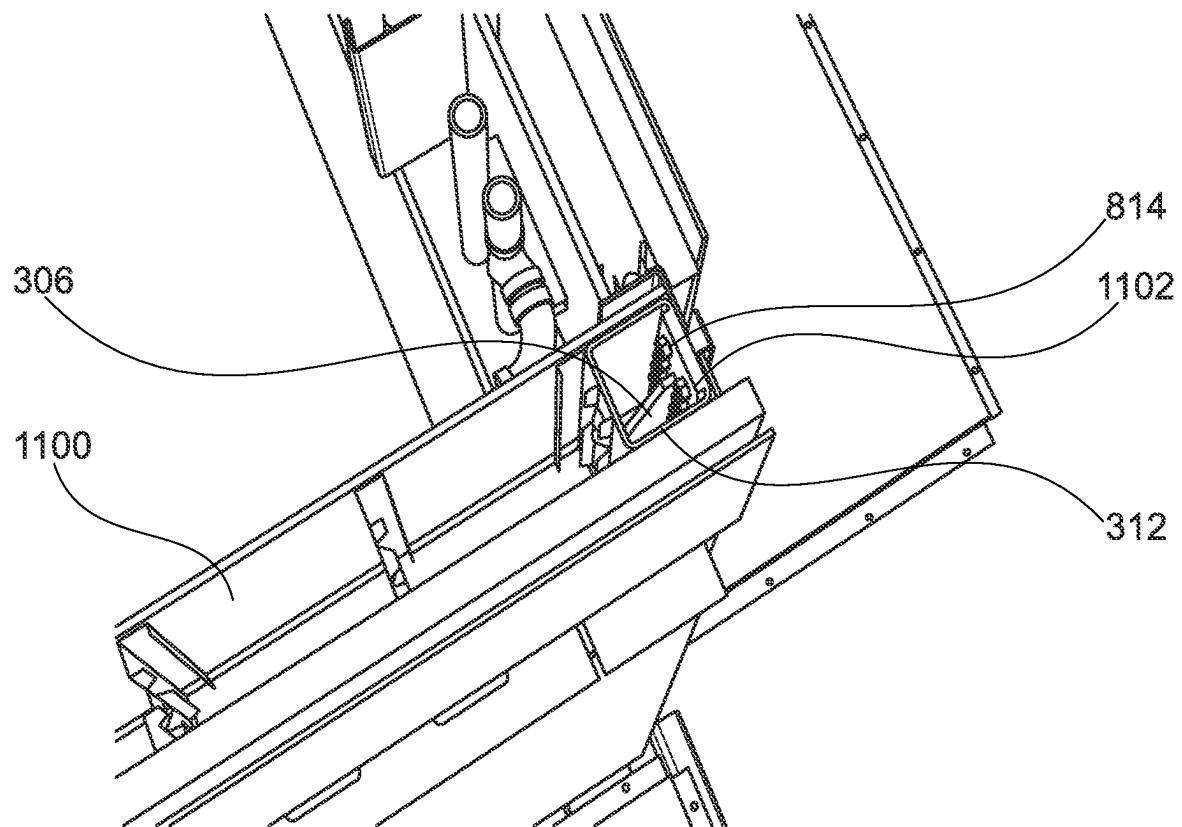


FIG. 11

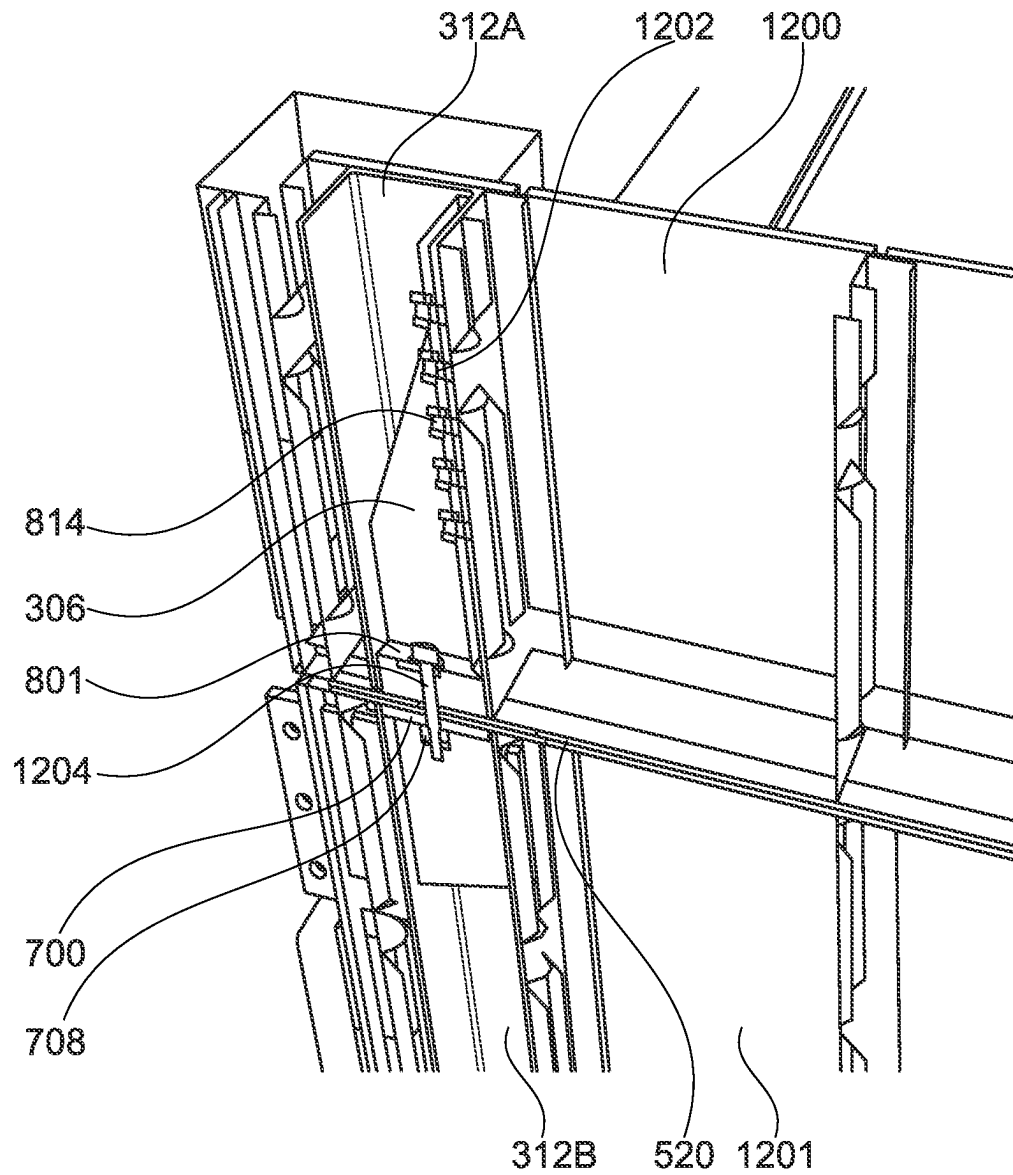


FIG. 12

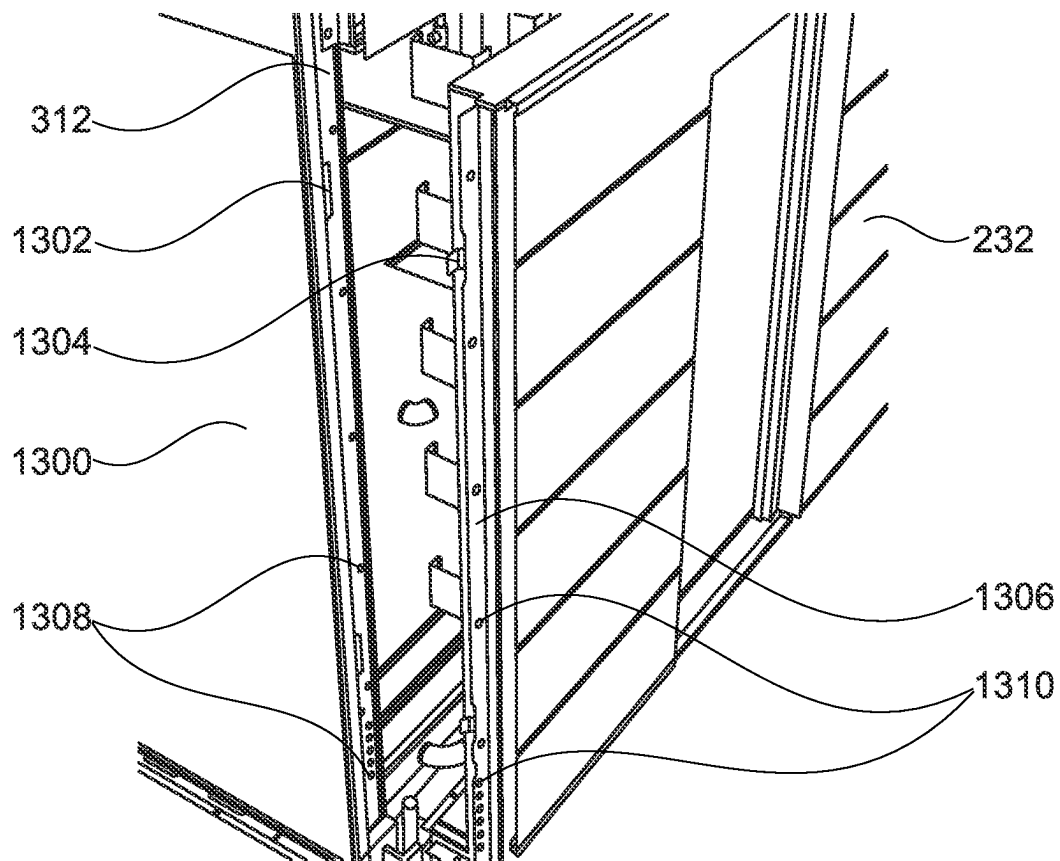


FIG. 13

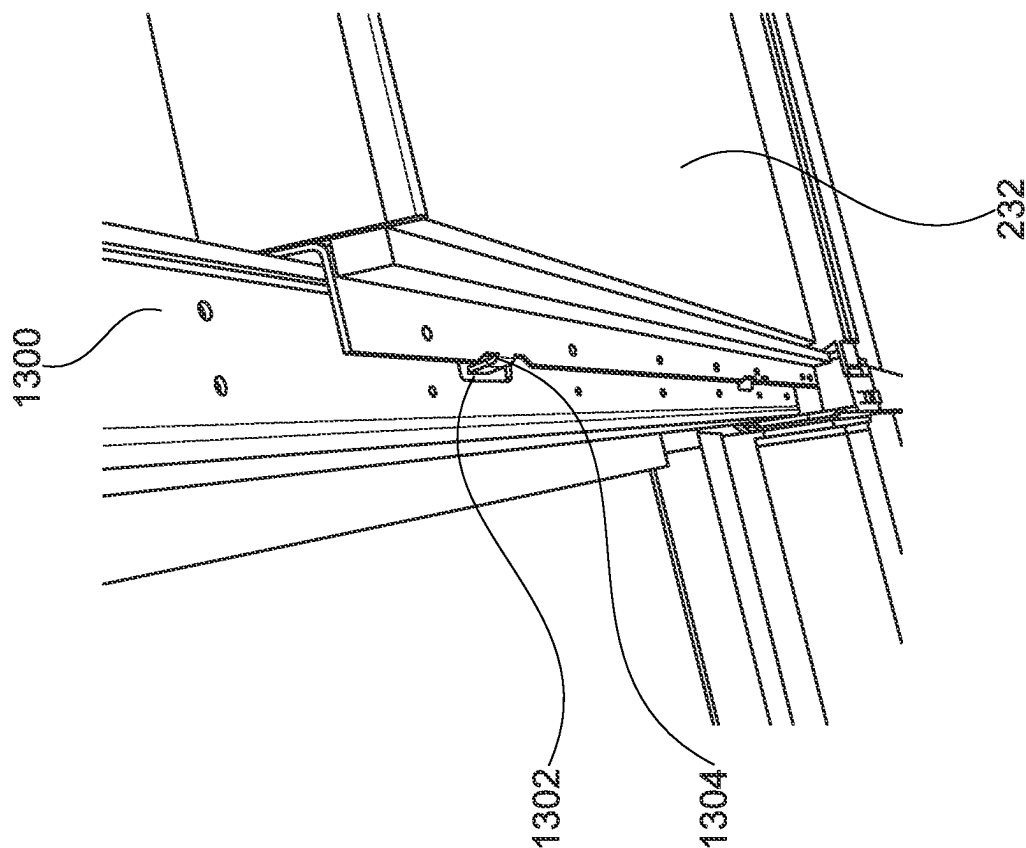


FIG. 15

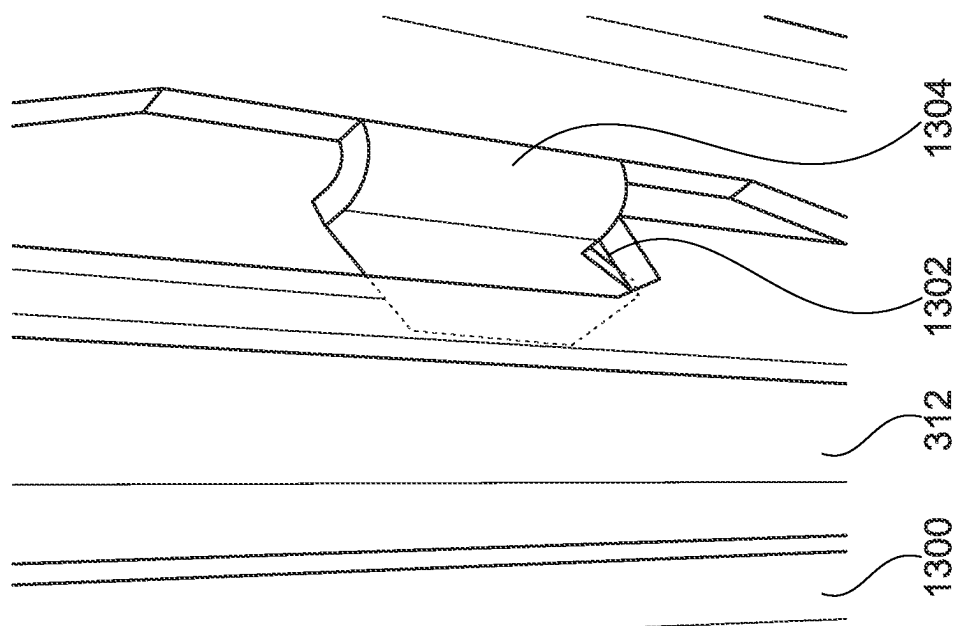


FIG. 14

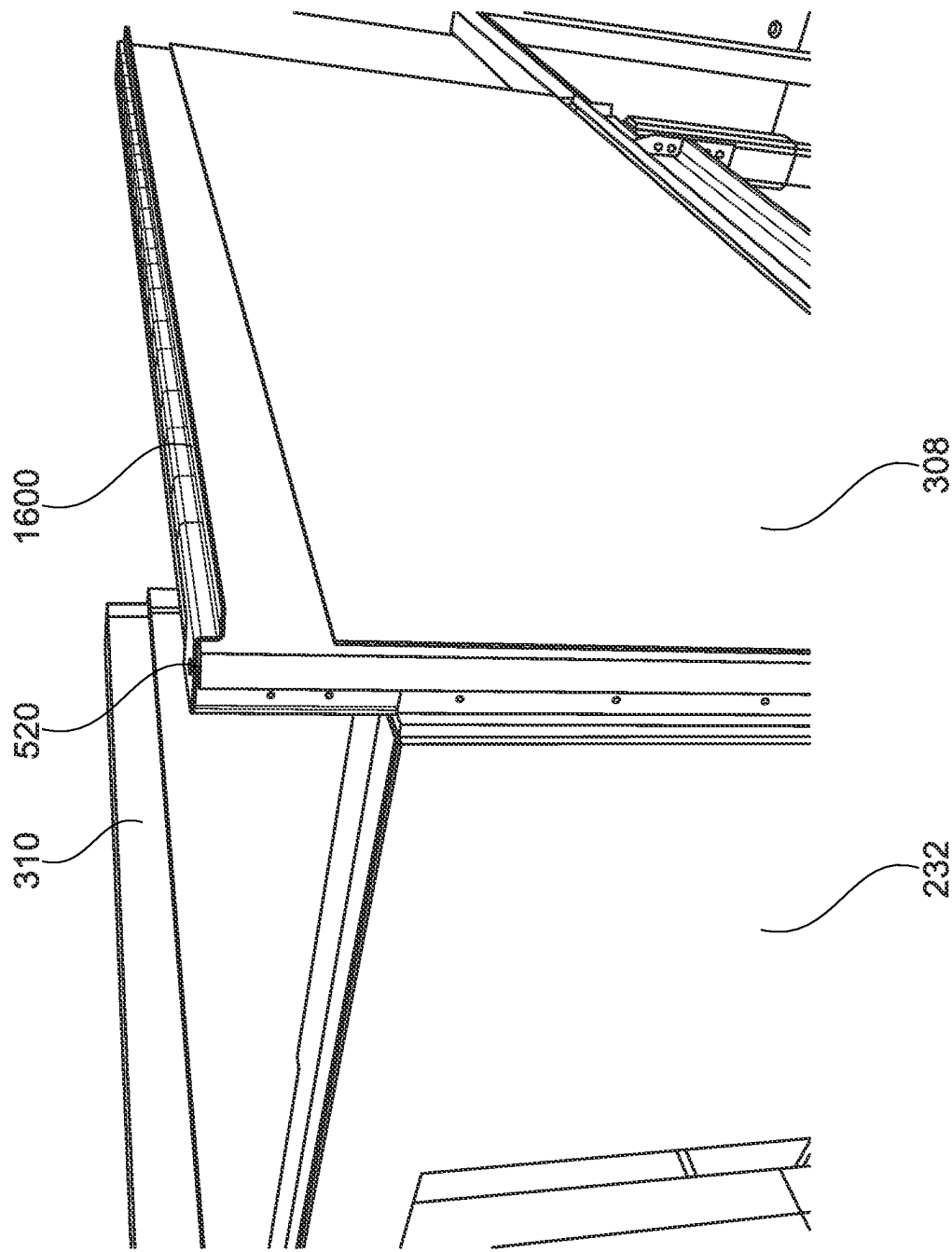


FIG. 16



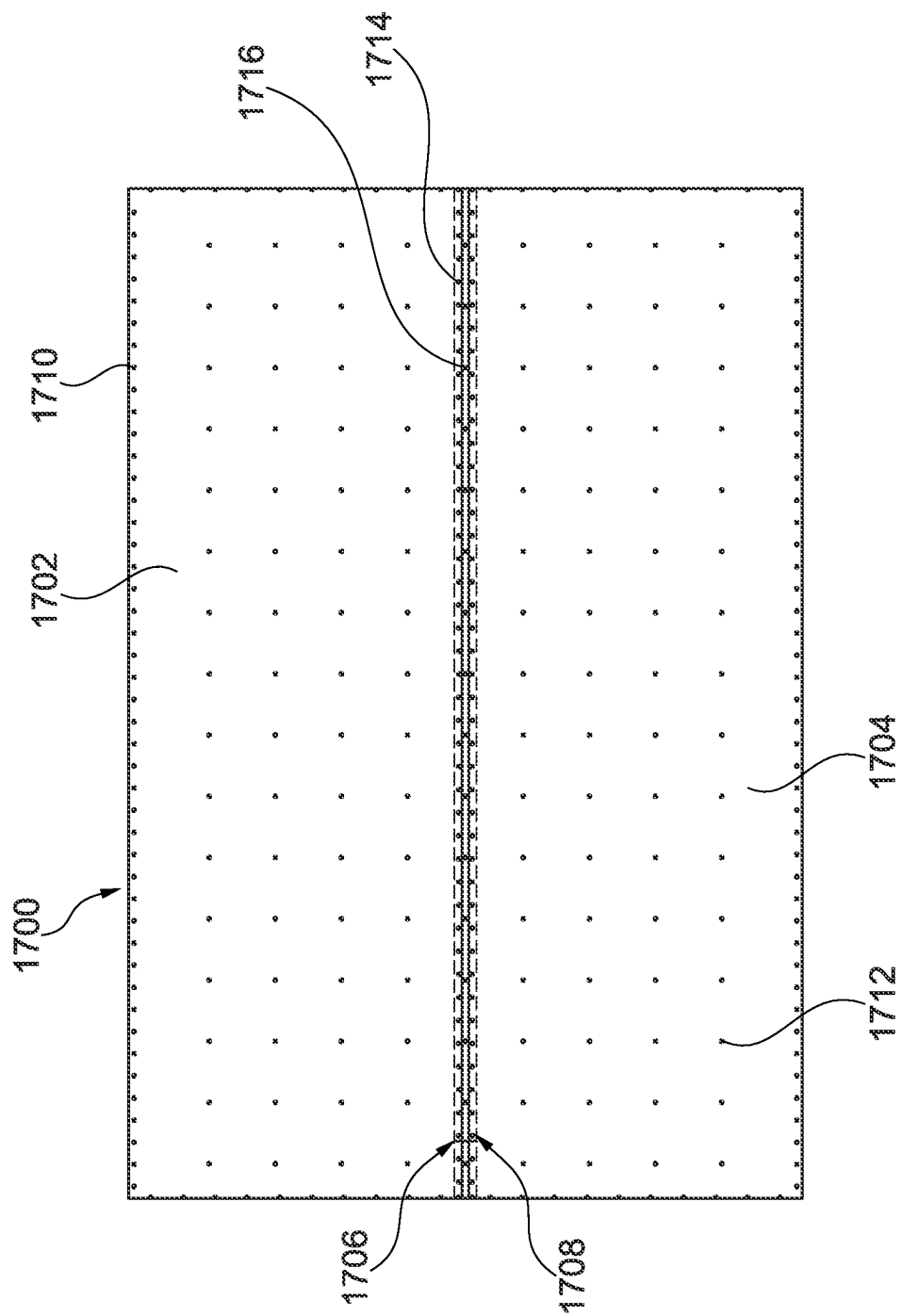


FIG. 17

**PRE-MANUFACTURED LOAD BEARING WALLS FOR A MULTI-STORY BUILDING****CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is the U.S. national phase filing under 35 U.S.C. § 371 of International Patent Application No. PCT/US2021/056077, filed Oct. 21, 2021, which claims priority under 35 U.S.C. § 119 (e) and/or under PCT Article 8 to U.S. Provisional Patent Application No. 63/104,239, filed on Oct. 22, 2020, and entitled "LOAD BEARING WALLS FOR A BUILDING" and to U.S. Provisional Patent Application No. 63/178,515, filed on Apr. 22, 2021, and entitled "LOW-MID RISE BUILDING HAVING LOAD BEARING WALLS, UTILITY WALLS, AND A CORRIDOR SYSTEM, AND OTHER ACCOMPANYING STRUCTURE, AND METHOD TO CONSTRUCT THE BUILDING." The contents of U.S. Provisional Patent Application Nos. 63/104,239 and 63/178,515 are incorporated herein by reference in their entirety.

The present application is related in subject matter to each of the following co-pending applications, each of which shares a common filing date of Oct. 21, 2021: International Patent Application No. PCT/US2021/056091, entitled "MULTI-STORY BUILDING HAVING LOAD BEARING WALLS AND METHOD TO CONSTRUCT THE BUILDING;" International Patent Application No. PCT/US2021/056074, entitled "MULTI-STORY BUILDING HAVING PODIUM LEVEL STEEL TRANSFER STRUCTURE;" International Patent Application No. PCT/US2021/056076, entitled "PRE-MANUFACTURED FLOOR-CEILING PANEL FOR A MULTI-STORY BUILDING HAVING LOAD BEARING WALLS;" International Patent Application No. PCT/US2021/056079, entitled "PRE-MANUFACTURED UTILITY WALL FOR A MULTI-STORY BUILDING HAVING LOAD BEARING WALLS;" International Patent Application No. PCT/US2021/056080, entitled "PRE-MANUFACTURED FLOOR-CEILING CORRIDOR PANEL FOR A MULTI-STORY BUILDING HAVING LOAD BEARING WALLS;" International Patent Application No. PCT/US2021/056081, entitled "MULTI-STORY BUILDING HAVING PREFABRICATED STAIR AND ELEVATOR MODULES;" and International Patent Application No. PCT/US2021/056086, entitled "PRE-MANUFACTURED FLOOR-CEILING DRAG ANCHOR FOR A MULTI-STORY BUILDING HAVING LOAD BEARING WALLS"; all of which are hereby incorporated by reference herein, in their respective entireties.

**BACKGROUND**

Conventional construction is typically conducted in the field at the building job site. People in various trades (e.g., carpenters, electricians, and plumbers) measure, cut, and install material as though each unit were one-of-a-kind. Furthermore, activities performed by the trades are arranged in a linear sequence. The result is a time-consuming process that increases the risk of waste, installation imperfections, and cost overruns.

Traditional building construction continues to be more and more expensive and more and more complex. Changing codes, changing environments, and new technology have all made the construction of a building more complex than it was 10 or more years ago. In addition, trade labor availability is being reduced significantly. As more and more craftsmen retire, fewer and fewer younger workers may be

choosing the construction industry as a career, leaving the construction industry largely lacking in skilled and able men and women to do the growing amount of construction work.

The construction industry is increasingly using modular construction techniques to improve efficiency. Modular construction techniques may include pre-manufacturing complete volumetric units (e.g., a stackable module) or one or more building components, such as wall panels, floor panels, and/or ceiling panels, offsite (e.g., in a factory or manufacturing facility), delivering the pre-manufactured modules or components to a building construction site, and assembling the pre-manufactured modules or components at the building construction site.

While modular construction techniques provide certain advantages over traditional construction techniques, challenges continue to exist in being able to meet housing and other building demands in communities. For example, the construction industry, whether using modular construction techniques or traditional construction techniques, needs to be able to address issues such as reducing construction costs and construction waste, reducing time to build, providing building designs that efficiently use space, and other challenges brought on by increasing demands for affordable housing and other building needs.

**SUMMARY**

An embodiment provides a pre-manufactured load bearing wall for a multi-story building. The load bearing wall includes:

- a plurality of parallel vertical metal studs;
- a first track affixed to an upper end of the studs
- a second track affixed to a lower end of the studs;
- a first vertical support member at a first edge of the wall;
- a second vertical support member at a second edge of the wall;
- a metal plate that runs between the first and second edges of the wall, and affixed to the first and second vertical support members and to the first track, wherein the plate is configured to support a horizontal section of a first angle of a floor-ceiling panel such that the floor-ceiling panel is hung from the load bearing wall;
- a horizontal member that runs between the first and second edges of the wall, that is affixed to the lower track, and that extends laterally away from the lower track, wherein the horizontal member is configured to be affixed to second angle of the floor-ceiling panel to provide a load path for a lateral load; and
- a stiffener device located at a top end of each of the first and second vertical support members, wherein the stiffener device is configured to provide rigidity for a joint between the first and second vertical support members and corresponding vertical support members respectively positioned serially above the first and second vertical support members.

Another embodiment provides a multi-story building. The building includes:

- a pre-manufactured floor-ceiling panel; and
- a pre-manufactured load bearing wall that includes:
  - a plurality of parallel vertical metal studs;
  - a first track affixed to an upper end of the studs
  - a second track affixed to a lower end of the studs;
  - a first vertical support member at a first edge of the wall;
  - a second vertical support member at a second edge of the wall;

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a metal plate that runs between the first and second edges of the wall, and affixed to the first and second vertical support members and to the first track, wherein the plate is configured to support a horizontal section of a first angle of the floor-ceiling panel such that the floor-ceiling panel is hung from the load bearing wall;

a horizontal member that runs between the first and second edges of the wall, that is affixed to the lower track, and that extends laterally away from the lower track, wherein the horizontal member is configured to be affixed to second angle of the floor-ceiling panel to provide a load path for a lateral load; and

a stiffener device located at a top end of each of the first and second vertical support members, wherein the stiffener device is configured to provide rigidity for a joint between the first and second vertical support members and corresponding vertical support members respectively positioned serially above the first and second vertical support members.

Still another embodiment provides method to manufacture a load bearing wall for a multi-story building. The method includes:

affixing a first track to an upper end of a plurality of parallel vertical metal studs;

affixing a second track to a lower end of the studs;

affixing a first vertical support member to a first outer stud at a first edge of the wall;

affixing a second vertical support member to a second outer stud at a second edge of the wall;

affixing a metal plate, which runs between the first and second edges of the wall, to the first and second vertical support members and to the first track, wherein the plate is configured to support a horizontal section of a first angle of a floor-ceiling panel such that the floor-ceiling panel is hung from the load bearing wall;

affixing a horizontal member, which runs between the first and second edges of the wall, to the lower track such that the horizontal member extends laterally away from the lower track, wherein the horizontal member is configured to be affixed to second angle of the floor-ceiling panel to provide a load path for a lateral load; and

placing a stiffener device at a top end of each of the first and second vertical support members, wherein the stiffener device is configured to provide rigidity for a joint between the first and second vertical support members and corresponding vertical support members respectively positioned serially above the first and second vertical support members.

#### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an illustration of an example multi-story building that can have pre-manufactured load bearing walls, floor-ceiling panels, and other building parts described herein, in accordance with some implementations.

FIG. 2 shows a partially constructed building having load bearing walls and floor-ceiling panels at a second floor level of the building, in accordance with some implementations.

FIG. 3 shows further details of the floor-ceiling panels and load bearing walls of the partially constructed building of FIG. 2, in accordance with some implementations.

FIG. 4 shows installation of a floor-ceiling panel onto load bearing walls for a next floor level of the building, in accordance with some implementations.

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FIG. 5 is a cutaway view that shows some of the components of load bearing walls in more detail, in accordance with some implementations.

FIG. 6 is a cutaway view of a load bearing wall, in accordance with some implementations.

FIG. 7 is another cutaway view of a load bearing wall, in accordance with some implementations.

FIG. 8 shows load bearing walls with spigots installed thereon, in accordance with some implementations.

FIGS. 9 and 10 show examples of protrusions for tubular members of load bearing walls and spigots attached to the tubular members, in accordance with some implementations.

FIG. 11 is a cutaway view showing parts of a load bearing wall that may be attached to a spigot, in accordance with various implementations.

FIG. 12 is a cutaway view showing parts of a load bearing wall that may be attached to a spigot, in accordance with various implementations.

FIG. 13 shows features of a load bearing wall to enable a utility wall to be hung from the load bearing wall.

FIGS. 14 and 15 show the load bearing wall of FIG. 13 having the utility wall hung therefrom.

FIG. 16 show a feature of a load bearing wall for supporting a corridor panel, in accordance with some implementations.

FIG. 17 is a front view of sheets of metal of a load bearing wall, in accordance with some implementations.

#### DETAILED DESCRIPTION

In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. The aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are contemplated herein.

This disclosure is drawn, inter alia, to methods, systems, products, devices, and/or apparatuses generally related to pre-manufactured load bearing walls that may be used in multi-story buildings having other pre-manufactured building parts (e.g., floor-ceiling panels, stair and elevator modules, steel transfer structures, corridor panels, etc.), such as a low-rise or mid-rise building. The load bearing walls are structural in that they are able to absorb and/or transfer lateral and/or vertical loads.

Traditionally, buildings are constructed using a steel structural frame that is designed to resist vertical and lateral loads. Thus, the structural frame can be thought of as a skeletal structure of a multi-story building, wherein the structural frame provides structural support for the building by absorbing vertical loads due to the weight of multiple stories and lateral loads such as due to wind or earthquakes, as well as providing the framing for various walls, floors, ceilings, and other components that can be affixed to the structural frame during the course of constructing the building. However, manufacturing and assembling such a traditional and extensive structural frame can be time consuming and costly in terms of labor and material. For instance, an

affordable housing crisis or other community needs may dictate that buildings with good structural integrity be built quickly and economically.

Therefore, various embodiments disclosed herein pertain to construction of a building using load bearing walls and other building parts such that the reliance upon a traditional structural frame can be reduced or eliminated, while at the same time enabling the building to meet lateral and vertical loading requirements. The load bearing walls can be pre-manufactured demising walls, end walls, or other vertical walls (including possibly utility walls), at least some of which are constructed and arranged so as to provide the structural support for the building in a manner that is sufficient to enable the building to handle vertical and lateral loads. The other building parts, such as the pre-manufactured floor-ceiling panels and corridor panels and their accompanying components, in combination with the load bearing walls and coupling linkages between them, also enhance the structural integrity for the building (e.g., for handling or transferring loads), improve acoustical performance, and increase fire safety.

The building may be a multi-story low-rise building or a multi-story mid-rise building in some embodiments. Each story of the building can include a single unit or multiple units. For instance, a particular unit may be living space, office space, retail space, storage space, or other human-occupied space or otherwise usable space in the building. In the context of living space, as an example, each story of the building may include multiple units to respectively accommodate multiple tenants.

The use of the pre-manufactured load bearing walls and other pre-manufactured parts enables the building to be constructed with a shorter time to build and at a lower cost (relative to a building that is constructed using a traditional structural frame), and without sacrificing the structural integrity of the building. Moreover, the floor-ceiling panels of the building may be made thinner relative to conventional floor-ceiling panels, thereby enabling the building to have more stories per vertical foot compared to a traditional building, or to have more open space per linear foot when relatively thinner load bearing walls are used. Thus, the building is able to provide more usable space (e.g., living space) as opposed to a traditional building that occupies the same footprint. In other cases, the thinner floor-ceiling panels provide more space between the floor and ceiling of each unit, which may be desirable for some occupants that prefer living spaces with “high ceilings.”

In some embodiments, the material composition of an entire module, as well as the wall, ceiling, and floor panels, may include steel. In some embodiments, the material composition may include aluminum. In still other embodiments, the wall, ceiling, and floor panels may be made from a variety of building suitable materials ranging from metals and/or metal alloys, composites, to wood and wood polymer composites (WPC), wood based products (lignin), other organic building materials (bamboo) to organic polymers (plastics), to hybrid materials, earthen materials such as ceramics, glass mat, gypsum, fiber cement, magnesium oxide, or any other suitable materials or combinations thereof. In some embodiments, cement, grout, or other pourable or moldable building materials may also be used. In other embodiments, any combination of suitable building material may be combined by using one building material for some elements of the entire module, as well as the wall, ceiling and floor panels, and other building materials for other elements of the entire module, as well as the wall, ceiling, and floor panels. Selection of any material may be

made from a reference of material options (such as those provided for in the International Building Code), or selected based on the knowledge of those of ordinary skill in the art when determining load bearing requirements for the structures to be built. Larger and/or taller structures may have greater physical strength requirements than smaller and/or shorter buildings. Adjustments in building materials to accommodate size of structure, load, and environmental stresses can determine optimal economical choices of building materials used for components in an entire module, as well as the wall, ceiling, and floor panels described herein. Availability of various building materials in different parts of the world may also affect selection of materials for building the system described herein. Adoption of the International Building Code or similar code may also affect choice of materials.

Any reference herein to “metal” includes any construction grade metals or metal alloys as may be suitable (such as steel) for fabrication and/or construction of the entire module, as well as wall, ceiling, and floor panels, and/or other components thereof described herein. Any reference to “wood” includes wood, wood laminated products, wood pressed products, wood polymer composites (WPCs), bamboo or bamboo related products, lignin products and any plant derived product, whether chemically treated, refined, processed or simply harvested from a plant. Any reference herein to “concrete” or “grout” includes any construction grade curable composite that includes cement, water, and a granular aggregate. Granular aggregates may include sand, gravel, polymers, ash and/or other minerals.

FIG. 1 is an illustration of an example multi-story building **100** that can have pre-manufactured floor-ceiling panels, load bearing walls, and other building parts (e.g., pre-manufactured corridor panels, utility walls, window walls, and other type of walls, etc.), in accordance with some implementations. It is noted that the building **100** of FIG. 1 is being shown and described herein as an example for purposes of providing context for the various embodiments in this disclosure. The various embodiments may be provided for buildings that have a different number of stories, footprint, size, shape, configuration, appearance, etc. than those shown for the building **100**.

The building **100** may be a multi-story building with one or more units (e.g., living, office, or other spaces) in each story. In the example of FIG. 1, the building **100** has six stories/levels/floors, labeled as levels L1-L6. Also as shown in FIG. 1, the building **100** has a generally rectangular footprint, although the various embodiments disclosed herein may be provided for buildings having footprints of some other shape/configuration. Moreover, each story may not necessarily have the same shape/configuration as the other stories. For instance in FIG. 1, level L6 of the building **100** has a smaller rectangular footprint relative to levels L1-L5.

The ground floor level L1 may contain living spaces, office spaces, retail spaces, storage spaces, common areas (such as a lobby), etc. or combination thereof. Levels L2-L6 may also contain living spaces, office spaces, retail spaces, storage spaces parking, storage, common areas, etc. or combination thereof. Such spaces may be defined by discrete units, separated from each other and from corridors or common areas by interior demising walls and utility walls (not shown in FIG. 1). An individual unit in turn may be made up of multiple rooms that may be defined by load bearing or non-load bearing walls. For example, a single unit on any given level may be occupied by a tenant, and may include a kitchen, living room, bathrooms, bedrooms, etc.

separated by walls, such as demising walls or utility walls. There may be multiple units (e.g., for multiple respective tenants) on each story, or only a single unit (e.g., for a single tenant) on a single story.

Each end of the building **100** includes an end wall **102**. One or more panels that make up the end wall **102** may span a single story in height. Any of the sides of the building **100** may include an end wall or a window wall **104** that accommodates a window **106**, such as window(s) for unit(s). One or more panels that make up the window wall **104** may span a single story in height. Some parts of the building **100** may include an end wall devoid of windows (e.g., not a window wall), such as an end wall **108**, which may be comprised of a panel that spans one story of the building **100**.

The unit(s) in each story may be formed using either an entire pre-manufactured module or from one or more pre-manufactured floor-ceiling panels (not shown in FIG. **1**) and wall panels, and the units may also adjoin each other via hallways having pre-manufactured corridor panels used as floor-ceiling panels. A floor-ceiling panel may form the floor of a first unit and a ceiling of a second unit below the first unit, and may also be used to form part of the roof of the building **100** when used as the ceiling panel for the top floor. The pre-manufactured wall panels may be used to form interior walls (e.g., demising walls, utility walls on a corridor, etc.), window walls (e.g., exterior window wall **104** that accommodate one or more windows **106**), utility walls (e.g., walls with utilities such as plumbing and electrical wiring contained therein), end walls, etc. According to various embodiments, at least some of these panels may be pre-manufactured off-site such as at a factory, and then installed on site by coupling them together to construct the building **100**. The various components of such panels and how such panels are attached to each other will be described later below.

The sides of interior walls that face the interior space (e.g., living space) of the building **100** may be covered by a finish panel, such as wall paneling, for decorative and/or functional purposes. Analogously, the tops and bottoms of floor-ceiling panels that face the interior space (e.g., living space) of the building **100** may also be covered with laminate flooring, finish panels, tile, painted/textured sheathing, etc. for decorative and/or functional purposes. For exterior walls such as end walls and window walls, the sides of these walls facing the outside environment may be covered with waterproofing membranes, tiles, glass, or other material for decorative and/or functional purposes.

According to various implementations, the building **100** is constructed using load bearing walls (such as demising walls, end walls, etc.). In this manner, such walls are able to support vertical loads, and non-shear walls are able to transfer lateral loads and shear walls are able to transfer and resist lateral loads. Because these walls are load bearing components, the building **100** can eliminate or reduce the use of an extensive steel structural frame in at least some of the levels. For instance, a steel structural frame (e.g., made of an array of beams and columns to which each and every floor-ceiling panel and wall are directly attached) may be absent in levels L2-L6. A steel structural frame may be used in level L1 and/or further structural reinforcement may be given to load bearing walls that are used in level L1 alternatively or in addition to a structural frame, so as to provide structural integrity at ground level.

The building **100**, having six levels L1-L6, is defined in some jurisdictions as a mid-rise building (e.g., buildings having six to 12 levels). Buildings having five levels and under are defined in some jurisdictions as a low-rise build-

ing. The various embodiments of the load bearing walls described herein may be used in low-rise and mid-rise buildings. Such low-rise and mid-rise buildings may have various fire ratings, with a 2-hour fire rating for mid-rise buildings of six stories or more and a 1-hour fire rating for buildings of five stories or less being examples for some of the buildings that use the load bearing walls described herein.

In some embodiments, the load bearing walls and other building parts described herein (in the absence of a structural frame, or with a reduced amount thereof) may be used for buildings that have a greater number of stories than a typical low-rise or mid-rise building. In such embodiments, the load bearing walls and/or other building parts described herein may be implemented with additional and/or modified structural components, so as to account for the increased load associated with the greater number of stories.

FIG. **2** shows a partially constructed building **200** having floor-ceiling panels **202** and load bearing walls **230** at a second floor level (L2) of the building, in accordance with some implementations. For purposes of example and illustration, the building **200** has a generally rectangular footprint, and is assumed to be a low-rise building having at most five stories (floor levels), and it is understood that the various implementations described herein may be used for buildings with other numbers of stories. A construction sequence described with respect to FIG. **2** and in the other figures that will be shown and described later may be adapted to construct buildings having other shapes, sizes, heights, configurations, number of stories, etc., such as the building **100** of FIG. **1** or any other building where load bearing walls, floor-ceiling panels, and the other building parts described herein are installed in the absence of extensive structural frames on at least some stories. In some embodiments, the various operations in the construction sequence may be performed in a different order, omitted, supplemented with other operations, modified, combined, performed in parallel, etc., relative to what is shown and described with respect to FIG. **2** and the other figures.

To describe a construction sequence to arrive at the partially constructed building **200** in FIG. **2**, a foundation **204** is first formed. The foundation **204** may be a steel reinforced concrete slab that is poured on the ground to define a footprint **206** of the building **200**, or may be some other type of shallow or deep foundation structure. Furthermore, excavation of the ground may also be performed to form a basement and/or elevator pit(s) **208** that form part of one or more elevator shafts to accommodate one or more elevators.

Next in the construction sequence, pre-manufactured stair and elevator modules **210** and **212** may be built on the foundation **204**, and positioned such that the elevator portions of the modules **210** and **212** that will contain the elevator shaft are superimposed over the elevator pit(s) **208**. The modules **210** and **212** according to various embodiments may be two stories in height, and there may be one or more of these modules per building, with two modules **210** and **212** shown by way of example in FIG. **2**.

Each of the modules **210** and **212** may be comprised of vertical columns made of steel, and horizontal beams spanning between the columns and also made of steel. Thus, the columns and the beams form a structural frame, which according to various embodiments is a load bearing structure that is able to withstand vertical and lateral loads. In other embodiments, the columns may be replaced by load bearing wall panels and the beams may remain as load bearing rings.

The modules **210** and **212** of various embodiments are positioned at specific locations of the foundation **204**. In the example of FIG. 2, the modules **210** and **212** are positioned on opposite sides of the building **200**. Other configurations may be used, such as positioning one or more modules at a central location in the building footprint **206** or at any other suitable location(s) on the building footprint **206**.

Next in the construction sequence, braced frames are installed on the foundation **204** in relation to the modules **210** and **212**. For example, braced frames **214** and **216** are arranged perpendicularly around and in close proximity to the module **210**, such that the module **210** is nested by the braced frames **214** and **216**. With respect to the module **212**, braced frames **218** and **220** are also arranged perpendicularly relative to each other but are spaced away from the module **212** by a greater distance.

The braced frames **214-220** may be arranged on the foundation **204** in any suitable location and orientation, dependent on factors such as the footprint or configuration of the building **200**, source of lateral and/or vertical loads, location/orientation for optimal stabilization, etc. Any suitable number of braced frames may be provided at the ground level. The braced frames may further vary in configuration. The example of FIG. 2 depicts some brace frames (e.g., the braced frame **218**) that are generally planar in shape (made of two columns and at least one horizontal beam that joins the two columns), with cross beams (X shaped beams) at the center of the braced frames. The braced frames **214-220** may span one, two, or other stories in height or intermediate heights, and multiple braced frames may also be vertically coupled.

According to various embodiments, the modules **210** and **212** are used as erection aids that guide the positioning and orientation of the braced frames **214-220** during construction. For instance, the modules **210** and **212** are installed first, and then the braced frames **214-220** are arranged relative to the location of the modules **210** and **212**. The braced frames may be directly welded (or otherwise attached/connected) to the modules, or may be linked to the module(s) over a distance via linking beams or other structural framing. In this manner, the modules **210** and **212** stabilize the braced frames **214-220**, and the braced frames **214-220** can operate to also absorb vertical and lateral loads from the building **200** via their linking connections.

The next phase of the construction sequence involves the erection of a steel transfer structure **222** (e.g., a podium structure) at ground level. The steel transfer structure **222** comprises a steel frame that receives and transfers load to the foundation **204** and to the braced frames **214-220**. The steel transfer structure **222** may have vertical members **224** (columns) having a height that spans one story, girders **226** that join pairs of columns **224**, and beams **228** that perpendicularly join pairs of girders **226**. The steel transfer structure **222** may further include vertically oriented “spigots” and/or other protrusions or engagement features to aid in construction, as will be described more fully below.

After completion of the steel transfer structure **222**, the next phase of the construction sequence involves the placement/installation of the floor-ceiling panels **202** over consecutive beams **228**, and more specifically, hanging the floor-ceiling panels **202** onto the beams **228**. A floor deck comprised of floor-ceiling panels **202** thus results after such installation.

Afterwards, the load bearing walls **230** (e.g., demising walls and end walls) are installed by being positioned over the beams **228**, and utility walls **232** are then installed by being hung onto the load bearing walls **230**. Next, corridor

panels **234** (which may be formed similarly in some respects as the floor-ceiling panels **202**) are hung from the utility walls **232**.

According to the example depicted in FIG. 2, the space between consecutive beams **228** and parallel girders **226** is sized to receive three adjoining floor-ceiling panels **202**, although the size of the floor-ceiling panels and the space between consecutive beams **228** and girders **226** can vary from one implementation to another. For instance, some implementations may install a multiple floor-ceiling panel between consecutive beams **228** that may vary in widths from 13 feet, to 16 feet, to 20 feet, to 24 feet, etc.

FIG. 3 shows further details of the floor-ceiling panels and load bearing walls of the partially constructed building **200** of FIG. 2. More particularly, FIG. 3 depicts the placement of three floor-ceiling panels **300A-300C** (collectively **300**) over and between consecutive beams **228A** and **228B** (collectively **228**). In the example shown, the floor-ceiling panel **300A** is adjacent to a window wall (not yet installed in FIG. 3) that faces an exterior of the building **200**, the floor-ceiling panel **300B** is adjacent to a utility wall (not yet installed in FIG. 3) that faces the interior corridor of the building **200**, and the floor-ceiling panel **300C** is a middle panel joined to and between the floor-ceiling panels **300A** and **300B**.

An installation sequence for the floor-ceiling panels may involve installing the floor-ceiling panel **300A**, floor-ceiling panel **300C**, and floor-ceiling panel **300B** in any suitable sequence. After these three floor-ceiling panels are installed, then the installation sequence moves to the next adjacent space between consecutive beams **228** (e.g., to the left direction in FIG. 3) so as to install the next three floor-ceiling panels in the same manner. This installation sequence repeats until all floor-ceiling panels are installed on the steel transfer structure **224** in the manner as depicted in FIG. 2 to complete a floor deck for that story.

FIG. 3 shows an example mounting of floor-ceiling panels, wherein if the north-south direction along the beam **228** is considered to be a transverse direction, and if the east-west direction along the girder **226** is considered to be the longitudinal direction, then the floor-ceiling panel **300** includes an angle **302** (or other piece of metal that provides ledge-like structure) that runs along its transverse direction along an upper surface (upper corner edge) of the floor-ceiling panel **300**. It is understood that the terms longitudinal and transverse are used as relative terms herein for the sake of convenience in describing perpendicular/orthogonal relationships between two components in the various embodiments, and may be swapped if the building **200** is being viewed or described from a different point of reference.

As will be shown and described in further detail below, the angle **302** includes a horizontal section that rests on a top surface of the beam **228A**. A vertical section of the angle **302** is attached to a vertical edge of the floor-ceiling panel **300**. A similar angle **302** is attached to the other/opposite transverse edge of the floor-ceiling panel **300**, and also has a horizontal section that rests on top of a beam **228B** adjacent to that edge of the floor-ceiling panel **300**. In this manner, the floor-ceiling panel **300** is hung by its transverse edges between two consecutive beams **228**.

With such an arrangement, the floor-ceiling panels **300** each provide a diaphragm that absorbs lateral and/or vertical load(s) and then transfers the load(s), via the angle **302**, to the beams **228** of the steel transfer structure **222** and/or to other supporting structure linked to the angles **302**. The steel transfer structure **222** then transfers the load(s) via one or

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more load paths to the foundation **204** and/or to the braced frames (e.g., the braced frames **214-220**) via connecting links.

According to some embodiments, the floor-ceiling panels **300** are supported between beams **228** along their transverse sides and are unsupported (e.g., by the girders **226**) along their longitudinal sides. Load bearing walls are positioned along and over the transverse sides/edges of the floor-ceiling panels **300**. As depicted in FIG. 3 by way of example and as will be described in further detail below, an end wall **308** (or a demising wall) may be positioned over/along a first transverse side/edge of the floor-ceiling panel **300**, and a demising wall **310** (or an end wall) may be positioned over/along a second transverse side/edge (opposite from the first transverse side/edge) of the floor-ceiling panel **300**.

Both of the walls **308** and **310** are load bearing walls. The end wall **308** is also a shear wall (but may not be a shear wall in some situations), and the demising wall **310** may or may not be a shear wall. In general, various structural configurations may be used to enable a wall to be a shear wall so as to resist in-plane shear and overturning. For example, stronger stud configurations or wall material may be used, as well as more dense screw patterns for attaching metal sheets to the walls and augmentation of vertical connections between panels at end studs (tubular members).

FIG. 3 also depicts alignment/placement and securing of the walls, using spigots **306** or other type of engagement/alignment feature. More particularly, the end wall **308** and the demising wall **310** are installed by initially positioning these walls over/above the beams **228**, and then lowering these walls so that they rest upon the beams **228**.

In the example of FIG. 3, the end wall **308** may include a tubular member **312**, such as a hollow structural section (HSS) tube, or other type of vertical support member affixed along both of its vertical proximal and distal edges. As the end wall **308** is being lowered into position, the spigots **306** (located adjacent to both ends of the beam **228**) are inserted into the openings of the lower ends of the tubular members **312**. The end wall **308** is then secured in place by tightening the attachment bolts on the spigot **306** and by affixing a lower edge of the end wall **308** to the upper surface of the floor-ceiling panels, which will be shown and described in further detail below with respect to FIG. 5.

A similar procedure may be used to install the demising wall **310**, by fitting openings (at lower ends of vertical tubular members **314** located at the vertical proximal and distal edges of the demising wall **310**) over and around the spigots **306**. A result of this installation is shown in FIG. 3, wherein two parallel walls are now standing in a self-aligned and self-supported manner, without the need for additional bracing from structural framing (e.g., an internal framing/skeleton of the building **200**).

After the floor-ceiling panels, walls, and corridor panels are finished being installed on the second floor level L2, then the construction sequence described with respect to FIGS. 2 and 3 repeats for each subsequent floor level above. For example, FIG. 4 shows installation of a floor-ceiling panel **400** on a next floor level of the building **200**, which in this example is the third floor level L3.

In FIG. 4, the floor-ceiling panel **400** is hung onto the previously installed end wall **308** and demising wall **310**, via the angles **302** that run along the transverse upper edges of the floor-ceiling panel **400**. The floor-ceiling panel **400** is hung by resting the horizontal sections of the angles **302** on the top surfaces of the end wall **308** and demising wall **310**. The manner in which the floor-ceiling panel **400** of FIG. 4 is hung from the top surfaces of the end wall **308** and the

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demising wall **310**, via the angles **302**, may be generally similar to the manner that the floor-ceiling panels **300** are hung from the top surfaces of the beams **228** in FIG. 3.

Holes **402** may be formed (e.g., offsite at the factory) in the angles **302** of the floor-ceiling panel **400**. During installation, such holes **402** may be superimposed over holes **410** formed at the upper surfaces of the walls **308/310**, so as to facilitate the alignment and positioning of the floor-ceiling panel **400** with some precision. For instance, temporary pegs or screws may be inserted into the holes **402** and **410** during installation to align and hold the floor-ceiling panel **400** in place, while the angle **302** is screwed, bolted, or welded to a top plate on the top surfaces of the walls **308** and **310**.

Moreover, further spigots **306** may be installed on top of the walls **308** and **310**, for alignment and securing of the upper end wall and demising wall that will be installed next on top of the respective lower end wall **308** and demising wall **310**. The angle **302** may have cutouts **404** to accommodate fasteners (e.g., bolts) for a mounting base of spigots **306** and/or to accommodate other parts or fasteners.

FIG. 4 also shows an installed utility wall **406** (e.g., the utility wall **232** shown in FIG. 2), with the floor-ceiling panel **400** being sized and installed such that the gap **304** is provided to accommodate a next utility wall above the utility wall **406**. The gap **304** may be absent in other embodiments. FIG. 4 further shows that the gap **304** accommodates utilities **408** (e.g., plumbing, electrical, etc.) that are pre-installed in the utility wall **406**. As will be described further below, the load bearing walls **308/310** may be constructed such that the utility wall **406** is hung from these load bearing walls.

FIG. 5 is a cutaway (or cross-sectional) view that shows some of the components of load bearing walls (e.g., demising walls such as the demising walls **310**) in more detail, in accordance with some implementations. More specifically, FIG. 5 shows a top portion of a load bearing wall (e.g., the demising wall **310**) and the bottom portion of a demising wall **500** that is positioned (on the next floor level) above the demising wall **310**. The top portions of both walls are the same/similar for both walls and the bottom portions of both walls are the same/similar for both walls, and so these walls will be described interchangeably herein.

The interior of each wall includes vertical parallel metal studs **510** along the entire length of the walls. The studs may be spaced at 18" on center (e.g., their centers are 18 inches apart), 16" on center, or some other spacing. Traditional studs are spaced at 24 inches on center, and so the reduced spacing distance between the studs **510** enables more studs to be present per linear foot of the length of the wall, thereby providing the wall with increased capability to support a vertical load.

According to some embodiments, each individual stud **510** may be formed from one or more vertically arranged C-channels having example dimensions of about 2 inches by 6 inches nominally, or 3½ inches by 1½ inches, with a 12 gauge, 14 gauge, or 16 gauge thickness (or other thinner or thicker gauge). One or more horizontally arranged C-channels may provide an upper track **512** (first track) attached/affixed to the upper ends of the studs **510**, and a lower track **514** (second track) attached/affixed to the lower ends of the studs **510**. The tracks **512/514** may be 14 gauge or other gauge/thickness of steel.

Each of the walls **310/500** may include a metal layer **516** on both sides of each wall **310/500**, such as sheet of metal made of 20 gauge steel or other steel gauge/thickness. The metal layer **516** covers the internal structure and other parts of the walls (such as the studs **510**, utilities, insulation, etc. in the interior of the walls, not shown in FIG. 5), and may

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be affixed to the studs **510** or tracks **512/514** by screws or other fasteners, welding, adhesives, etc. A layer **518** may overlie each respective metal layer **516**, and may comprise a gypsum board, a finish panel, etc.

A horizontal metal plate **520** may be affixed on top of the upper track **512**, and may run along the entire length of the upper edge of the lower wall **310** or along a portion thereof. The metal plate **520** may be steel, for example, that is about 7 inches wide and ¼ inches thick. The metal plate **520** may be affixed (offsite at a factory) to the upper track **512** via screws or bolts or other fasteners, by welding to the upper track **512** and/or to the metal layer **516**, or by some other attachment method.

The lower portion of each wall **310/500** has an affixed horizontal member **522** (e.g., welded to the lower track **514** and/or to the metal layer **516**, bolted or screwed to the lower track **514**, etc., which may be performed offsite at a factory). The horizontal member **522** may run along an entire length of the lower track **514** (e.g., runs along the full length between the proximal and distal ends of the wall **500**), or may run intermittently in sections along the lower track **514**, such as depicted in the example of FIG. 5. The horizontal member **522** extends laterally away from the lower track.

According to various implementations, the horizontal member **522** may be made from 14 gauge steel, or ¼ inch steel, or some other steel gauge or thickness. The horizontal member **522** may be 12 inches wide or other dimension. It is understood that the foregoing various dimensions (as well as for various other components described throughout this disclosure), such as thicknesses, gauges, lengths, widths, heights, etc. are for illustrative purposes, and that such dimensions may vary from one implementation to another depending on factors such as material availability, cost considerations, structural performance requirements (including loading and weight requirements), design variations, etc.

A floor-ceiling panel **400A** and an adjacent floor-ceiling panel **400B** (collectively **400**) are hung onto a load bearing demising wall (e.g., the demising wall **310** of FIG. 5), via the respective horizontal sections **502A** and **502B** (collectively **502**) of the angles **302A** and **302B** (collectively **302**) of the respective floor-ceiling panels **400A** and **400B**.

The angle **302** (e.g., an L-shaped member such as a hot-rolled metal angle or other type of load carrying/bearing angle) is positioned over and rests on the plate **520** serving as a head plate at the top of the wall **310**. The angle **302** (first angle) includes, in addition to the horizontal section (flange) **502**, a vertical section (flange) **504**.

The floor-ceiling panel **400** further includes a shear angle **506** (which may also be a cold formed metal angle of 14 gauge, for example) that runs along each of the transverse upper edges/corners of the floor-ceiling panel **400**. The shear angle **506** (a second angle) has a horizontal section **508** and a vertical section **528**. The horizontal section **508** of the shear angle **506** lies on top of and may be attached to an upper surface of the floor-ceiling panel **400**. The angle **302** may be ¼ inch steel, with vertical and horizontal sections of lengths between 2-6 inches, for example. The shear angle **506** may be the same dimensions as the angle **302**, or may be made of relatively thinner (or thicker) steel with horizontal/vertical sections that are shorter (or longer) relative to the angle **302**.

During offsite manufacturing, the vertical section **528** of the shear angle **506** is welded to the vertical section **504** of the angle **302** (such as via a continuous weld or a stitch weld) at the upper edge of the vertical section **528**. The vertical section **504** of the angle **302** is then welded (such as via a

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continuous weld or a stitch weld), during the off-site manufacturing, to an end member **524** (such as a track in the form of a C-channel) attached to the ends of longitudinally running parallel metal joists **526** of the floor-ceiling panel **400**.

Welding or otherwise attaching the vertical section **504** of the angle **302** to the floor-ceiling panel **400** enables the angle **302**, after being hung to the wall **310** during the construction sequence, to support the vertical load of the floor-ceiling panel **400**. The horizontal section **508** of the shear angle **506** in combination with the angle **302** also provides a load path to enable lateral load to be transferred from the diaphragm, formed by the floor-ceiling panel **400**, to the plate **500** and then to load path(s) or connecting links to the braced frames, etc.

This arrangement of the shear angle **506** and the angle **302** thus forms a T-shaped element that each run transversely along the entire length of the upper corner edges of the floor-ceiling panel **400**. While the examples are described herein of the shear angle **506** and the angle **302** (both made of metal such as steel) being separate pieces that are attached to each other, some embodiments may use a single integrated piece of metal that is T-shaped.

During the construction sequence and after the floor-ceiling panels **400** are hung from the wall **310**, the wall **500** is lowered aligned/positioned into place (e.g., using the spigots **306** as previously explained above), and then the horizontal member **522** is affixed to the each of the horizontal sections **508** of the shear angles **506** of the floor-ceiling panels **400**, thereby permanently mounting the upper load bearing wall **500** over the lower load bearing wall **310**. The horizontal section **508** thus forms a landing/fastening location for the horizontal member **522**. The horizontal member **522** may be affixed to the shear angle **506** such as by screwing, by stitch or continuous welding the horizontal member **522** to the horizontal section **508**, or by bolting or other attachment technique.

Affixing the load bearing upper wall **500** to the floor-ceiling panels **400** in this manner enables lateral load to transfer from the upper wall **500** to the horizontal member **522**, and then to the shear angle **506** and/or the angle **302**. The lateral load can then transfer across the diaphragm formed by the floor-ceiling panel **400** (e.g., via the sheets of steel in the floor-ceiling panel **400**) and then to linking connections with further load bearing walls (e.g., via other angles **502/506** and plates **500**), other floor-ceiling panels, corridor panels, and through various other linking elements and other possible load paths, and then to resisting elements such as the braced frames, designated shear walls, etc. For example, in the case that the depicted walls are shear walls, lateral forces may follow the path **500** to **522** to **506/302** to **520**, and down to wall **310**, thereby transmitting collected lateral force from the diaphragm down the shear wall to the steel transfer structure at ground level and into the foundation. These lateral forces may include forces from non-shear bearing walls that are transmitted into the diaphragm by the same connection detail (as described).

FIG. 6 is a cutaway view of a load bearing wall (e.g., one of the demising walls **310** of FIG. 3 or **500** in FIG. 5), in accordance with some implementations. The upper track **512** and the plate **520** have been cut away from FIG. 6, for further clarity. The wall **310** is shown in FIG. 6 as being erected, in a generally similar manner as shown in FIG. 3 and the other figures.

The studs **510** are shown, with the metal layers **516** respectively attached to both sides of the studs **510**. Other



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parts of (or attachments) to the wall **310**, such as the layers **518** and a fascia **602**, are also shown in FIG. **6** for context.

According to various embodiments, the load bearing wall (e.g., the demising wall **310** or the end wall **308**) includes a vertically running tubular member **312** (also shown in FIG. **3** and which may also be a similar/same type of part as the tubular member **314**) at both the proximal end and the distal end of the wall **310**. The tubular member **312** may be, for example, a single HSS tube having dimensions of 6 inches by 6 inches by  $\frac{1}{4}$  inch thick. HSS tubes of other dimensions may be used for the tubular member **312**, including bundles of two or more HSS tubes that are attached (e.g., by welding) together.

In other implementations, two or more C-channels, I-beams, box beams, or other type of elongated vertical support member may be bundled together to provide the structural support analogous to that provided by the HSS tube shown in FIG. **6**, at the proximal and distal ends of the wall **310**. Such other type of support member also may not necessarily be tubular in form, individually and/or after being bundled together. For the sake of explanation, the various embodiments herein will describe the vertical support member at the proximal and distal ends of the wall as being the tubular member **312** (and similarly the tubular member **314**) such as an HSS tube.

The tubular member **312** is attached, offsite at a factory, to an outer stud **604**. This attachment may be done by welding together the tubular member **312** and the outer stud **604**, or via the use of fasteners such as bolts/nuts or screws.

FIG. **7** is another cutaway view of a load bearing wall (e.g., the demising wall **310** of FIG. **6**), in accordance with some implementations. In this view, the plate **520** is now shown as being attached to the upper portion of the wall **310**. Also shown are the hole(s) **410** (described with respect to FIG. **4** above) that run through the plate **520**, and which is a temporary construction fastening hole for use with a corresponding hole in the angle **302**. The connection at **410** holds floor-ceiling panel in place for precision and safety during building erection. This fastening also creates a tight joint for weld setup.

The tubular member **312** includes a cap **700** that is sized and shaped to cover the top opening of the tubular member **312**. The cap **700** may be the same thickness as the walls of the tubular member **312** (e.g.,  $\frac{1}{4}$  inches thick), or may be thinner or thicker.

A vertical slot **702** may be formed at the upper end of the tubular member **312**, which is sized and shaped to accommodate a protrusion **704**. The protrusion **704** may be oriented in a vertical direction, so as to be orthogonal/perpendicular to the cap **700** that is oriented in a horizontal direction. The protrusion **704** may pass through the slot **702**, so as to have a length that extends at least up to the outer surface of the tubular member **312**. In the example of FIG. **4**, the protrusion **704** extends past the slot **702** by a sufficient distance so as to form a flag-like structure.

According to various embodiments, the protrusion **704** in combination with the cap **700** operate as a stiffener device for the upper end of the tubular member **312**, such as for resisting seismic overturning loads when the wall is a shear wall. In implementations wherein the extended flag-like structure of the protrusion **704** is present, holes **706** may be provided in the protrusion **706** for coupling (e.g., via bolts) to other structural elements, such as a structural link across a corridor, framing for a balcony, or other structure, for purposes of connection and providing a load path for shear or vertical loads.

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In some embodiments, the protrusion **704** and the cap **700** may be formed from a single piece of metal that is vertically inserted into the opening of the tubular member **312**. In other embodiments, the protrusion **704** and the cap **700** may be separate pieces of metal that are joined together (e.g., by welding off site at a factory) and then vertically inserted into the opening of the tubular member **312**. Once inserted into place, the plate **520** is then installed (e.g., by welding or fasteners) over both the upper track **512** and the cap **700**.

In some embodiments, the cap **700** and/or the protrusion **704** may be welded to outer surfaces of the tubular member **312**, prior to attachment of the plate **520**, so as to hold the cap **700** and protrusion **700** in place. In other embodiments the cap **700** (with the protrusion **704** attached thereto) may be loose-fitted over the opening of the tubular member **312** and into the slot **704**, and then held in place when the overlying plate **520** is fixedly attached to the upper surface of the wall **310**.

Still further, an underside of the cap **700** may be provided with a captive fastener (such as a captive nut **708**) to receive a bolt (not shown in FIG. **7**) that is passed through a hole **710** that runs through the plate **520** and cap **700**. Such a bolt may, for example, serve to attach a spigot **306** to the plate **520** in a manner that will be described next below, and at the same time, to also firmly hold the cap **700** in place against the plate **520** and the spigot **306**. This bolt connection in conjunction with connections at the protrusion **706** allows the plate **520** (and by extension the end wall) to be used as a linking member to transfer lateral forces. Other holes **712** may be formed in the plate **520** to receive bolts for the spigot **306** and/or for connection of other parts and/or for alignment.

FIG. **8** shows load bearing walls with spigots **306** installed thereon, in accordance with some implementations. For example, load bearing walls are erected in FIG. **8**, and may include an end wall **800** (similar to the end wall **308**) and a demising wall **802** (similar to the demising wall **310**). A utility wall **232** is hung from the end wall **800** and the demising wall **802**.

Respective spigots **306A** and **306B** (collectively **306**) are shaped and sized to fit inside tubular members **312/314** (shown in FIG. **3**) of the load bearing walls, and are shown in FIG. **8** as being installed on top of the walls **800** and **802**, so that the next (upper walls) can be fitted over the spigots **306**. Each spigot **306** includes a horizontal mounting base **801** (engaged, over the plate **520** of the wall and a shim, with the plate **700** via a nut at hole **710** in FIG. **7**), a vertical attachment section **804** that is orthogonal to the mounting base **801**, and a vertical bracket **806** that is orthogonal to the attachment section **804**. The bracket **806**, in combination with the attachment section **804** and the mounting base **801**, provide rigidity to enable each spigot **306** to operate as a type of anchor point to stabilize and align a next upper load bearing wall, when the tubular member **312/314** of that wall is inserted over and around the spigot **306**. For example, the spigots **306** attached via the attachment section **804** to the proximal and distal ends of an upper load bearing wall prevent in-plane shear and overturning of that load bearing wall, after that load bearing wall is firmly affixed over the spigots **306**. In some implementations, the bracket **810** may not be present in some spigots **306**, for example if the wall that overlies the spigot is expected to be subject to relatively weaker loads, and so less robust bracing of the spigot is needed.

The attachment section **804** of the spigot **306** further includes at least one alignment hole **808** to align with a respective hole in the tubular member **312/314** of a wall that

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will be installed over the spigot **306** and over the previously erected lower wall below, plus a plurality of holes **810** to receive bolts for securing that upper wall to the spigot **306**. Where the wall is a non-shear demising wall or other type of non-shear wall that may not experience significant shear forces (such as another demising wall positioned above the demising wall **802**), the spigot **306B** may be provided with a relatively smaller number of holes **810** to receive bolts that secure that upper wall to the spigot **306B**. Where the wall is an end wall or other type of wall that may experience significant shear forces (such as another end wall positioned above the end wall **800**), the spigot **306A** may be provided with a relatively higher number of holes **812** to receive bolts that secure that upper wall to the spigot **306A**. The higher number of holes correspond to a higher number of bolts to provide rigidity/strength to ensure that the wall does not twist or overturn when subjected to load, including shear forces. For the holes **810** and **812**, the spigots **306** may be provided with captive nuts **814**, since the nut-side of the holes may be inaccessible after the tubular members **312/314** of the upper walls are lowered over the spigots **306**.

FIGS. **9** and **10** show examples of protrusions **704** for tubular members of load bearing walls and spigots **306** attached to the tubular members, in accordance with some implementations. Such protrusions **704** were previously discussed above with respect to FIG. **7**.

In the implementation of FIG. **9**, the protrusion **704** for a load bearing wall **900** does not significantly extend past the outer surface of the tubular member **312**. This implementation may be used, for example, The stiffener plate shown in FIG. **9** may be implemented specifically where a demising wall is designed to act as a shear wall and the tube at each end of the wall encounters higher axial seismic overturning loads consistent with the use of a spigot type such as shown at spigot **306A** in FIG. **8**. In cases of a demising wall subject to lower loading (non-shear load bearing wall), the protrusion **704** as a stiffener may be omitted completely.

FIG. **9** further shows the spigot **306** having its mounting base **801** attached, via bolt **902**, through the plate **520** of the wall **900** to the plate **700**. The parts of the wall **900** that enable this attachment were described above with respect to FIG. **7**, and will be further described in FIG. **12**. The parts of a load bearing wall (not shown in FIG. **9**) above the wall **900** that will enable attachment of that upper wall to the attachment section **804** of the spigot **306** will be further described in FIGS. **11** and **12**.

FIG. **10** is analogous to FIG. **9**, except that the protrusion **704** extends a further distance from the surface of the tubular member **312** of a wall **1000**, thereby forming a flag-like structure. The flag-like structure of the protrusion **704** may thus be used to connect to other structural links, for purposes of carrying and transferring load, providing structural continuity and support, etc. As with FIG. **9**, a bolt **1002** may be used to affix the mounting base **801** of the spigot **306** to the plate **520** of the wall **1000**.

FIG. **11** is a top cutaway view showing parts of a load bearing wall **1100** that may be attached to a spigot, in accordance with various implementations. More specifically, an upper load bearing wall **1100** (generally structured in the similar/same manner as the other load bearing walls previously described above), such as an end wall, has been lowered into place over a lower load bearing wall, such that the spigot **306** has been inserted into the tubular member **312** of the wall **1100**.

As shown in FIG. **11**, the inside of the tubular member **312** has a generally rectangular profile that is sized and shaped to accommodate a generally rectangular footprint/

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profile of the spigot **306**. Furthermore, one side of the tubular member **312** has holes at location **1102** to enable bolts (not shown) to be inserted through these holes so as to engage the captive nuts **814** of the spigot **306**. This engagement of the bolts with the captive nuts **814** enables the spigot **306** (present on both proximal and distal ends of the wall **1100**) to align and secure the wall **1100**, including stabilizing the wall **1100** against twisting and overturn.

In the implementation of FIG. **11**, wherein the side of the tubular member **312** (at location **1102**) is exposed, the bolts may be inserted into the holes exposed at the exterior facing side of the tubular member **312** at the location **1102**, and tightened using wrenches or other tools before a fascia or other exterior layer covers the location **1102**. FIG. **12** shows another implementation, wherein holes in the tubular member **312** may be disposed at a different location.

For example, FIG. **12** is a side cutaway view showing parts of a load bearing wall **1200** that may be attached to a spigot **306**, in accordance with various implementations, wherein holes **1202** of the tubular member **312A** are disposed at an interior facing side of the tubular member **312**. With this implementation, bolts may be inserted from the interior of the wall **1200** through the holes **1202** so as to engage with the captive nuts **814** of the spigot **306**. An access panel may be provided at the wall **1200** in order to enable wrench/tool access to the bolts being inserted from the interior of the wall **1200**.

Moreover, FIG. **12** shows a bolt **1204** running through a hole in the mounting base **801** of the spigot **306**. The bolt **1204** then runs through the plate **520** of a lower wall **1201** underneath the wall **1200**, and then through the cap **700** of the tubular member **312B** of the wall **1201** so as to engage with the captive nut **708**. When tightened (e.g., on site with a tool prior to tubular member **312A** of the upper wall **1200** being lowered into position around the spigot **306**), the bolt **1204** stiffens the connection between the spigot **306** and the tubular member **312B**. Thus, when there are multiple spigots **306** arranged vertically between and that join together serially/vertically positioned tubular members **312** (in combination with the caps **700** and the protrusions **704** adjacent to the spigots **306**), a result is stiffer joints along the vertical direction and providing a feature by which shear walls resist axial overturning forces during a seismic event. Also, the tightened bolts **1204** provide additional tension between the tubular members **312** to further securely hold the walls in place.

FIG. **13** shows features of a load bearing wall **1300** to enable a utility wall **232** to be hung from the load bearing wall **1300**. More specifically, the tubular member **312** (or **314**) of the wall **1300** is formed with a plurality of slots **1302**. The slots **1302** are sized and shaped to receive a corresponding plurality of tabs **1304** of the utility wall **232**.

According to various embodiments, the utility wall **232** includes an angle **1306** that runs along both of its vertical edges. The angle **1306** includes or is formed with the plurality of tabs **1304** that fits into the slots **1302** of the wall **1300**.

The tabs **1304** may have any suitable shape. For example, the tabs **1304** may have a tapered shape so as to be more easily inserted into the slots **1302**. The tabs **1304** may also have a hook-shaped configuration in some implementations, so as to provide more secure placement. In still other implementations, the tabs **1304** may be located on the wall **1300**, and the slots **1302** may be located on the utility wall **232**.

Further attachment mechanisms may be used to hold the utility wall **232** in place. For instance, the tubular member

**312** may be provided with a plurality of holes **1308**, some of which may be alignment holes and some of which may be holes to receive fasteners (such as screws or bolts) that are inserted into corresponding holes **1310** formed in the angle **1306** of the utility wall **232**, thereby further securely attaching the utility wall **232** to the wall **1300**.

FIGS. **14** and **15** show the load bearing wall **1300** of FIG. **13** having the utility wall **232** hung therefrom. In FIG. **14**, the tab **1304** has a downward hook shape, such that when the tab **1304** is inserted into the slot **1302** of the tubular member **312** of the wall **1300**, the weight of the utility wall **232** hooks the tab **1304** firmly against the lower edge of the slot **1302**. FIG. **15** shows the utility wall **232** in its fitted/hung position against the wall **1300**.

FIG. **16** show a feature of a load bearing wall (e.g., the end wall **308**) for supporting a corridor panel located on the next floor level above, in accordance with some implementations. More specifically, a z-plate **1600** has a first leg positioned horizontally over the plate **520** of the wall **308**. The z-plate **1600** may run continuously or in sections along the upper surface of the wall **308**, from the proximal end to the distal end of the wall **308**.

A second leg of the z-plate **1600** projects outwardly from the face of the wall **308**, so as to form a ledge-like support structure. The ledge-like support structure of the z-plate **1600** may be used to support an angle of a corridor panel (a floor-ceiling panel running along a hallway/corridor), in a manner somewhat analogous to the plate **520** of the wall **310** in FIG. **3** supporting the horizontal section **502** of the angle **302** of the floor-ceiling panel **400**.

FIG. **17** is a front view of sheets of metal (e.g., the metal layer **516** of FIG. **5** and shown in the other previous figures) of a load bearing wall **1700**, in accordance with some implementations. The sheets comprise a first sheet **1702** that runs along an upper region of the wall **1700** and a second sheet **1704** that runs along a lower region of the wall **1700**. A metal strip **1706** underlies a seam **1708** between the first sheet **1702** and the second sheet **1704**.

The first sheet **1702** and the second sheet **1704** are attached (along a perimeter of the wall **1700**) to the underlying upper and lower tracks and to the tubular members of the wall **1700**, via a first set of screws **1710**. The screws **1710** may be relatively closely spaced, for example 2 inches on center, so as to provide structural strength and rigidity to the diaphragm formed by the first sheet **1702** and the second sheet **1704**.

A second set of screws **1712** may be used to attach the first sheet **1702** and the second sheet **1704** to the vertically running parallel studs **510** of the wall **1700**. The second set of screws **1712** may be spaced apart at a relatively longer spacing, such as 6 inches or 12 inches on center.

The first sheet **1702** and the second sheet **1704** may be attached to the metal strip **1706** using a third set of screws **1714** that run horizontally above and below the seam **1708**. The spacing between the third set of screws may be, for example, 2 inches on center. Furthermore, the metal strip **1706** provides a landing location for screws **1716**, at the seam **1708**, so as to enable the metal strip **1706** to be attached to the studs **510** of the wall **1700**.

According to various embodiments, the first sheet **1702** and the second sheet **1704** (as well as the metal strip **1706**) are fastened by these various screws only to the studs **510** and upper and lower tracks, and affixed to the tubular members via welds adjacent to studs. Then, any other layer that overlies the first sheet **1702** and the second sheet **1704**, such as gypsum boards or other exterior layers, are fastened (via screws) only to the first sheet **1702** and the second sheet

**1704**, and not to the studs **510**, tracks, and tubular members. This discontinuity provides improved acoustical performance (e.g., sound proofing).

Moreover, the first sheet **1702** and the second sheet **1704** also improve the fire rating and acoustical separation between occupiable space of the building, and provide an air barrier between interior and exterior climates. Pre-installed electrical and/or plumbing utilities can comprise parts of the load bearing wall **1700** and other load bearing walls described herein, and may overlie the first sheet **1702** and the second sheet **1704**.

The first sheet **1702** and the second sheet **1704** may be 20 gauge in thickness, for example. The metal strip **1706** may be flat stock that is 4 inches wide and 18 gauge in thickness, for example.

According to various embodiments, the gauges/dimensions/configuration of the studs and tubular members, of all load bearing walls in the building **100**, may be generally uniform from one wall to another wall, whether a shear wall versus a demising wall that is not a shear wall and/or whether an upper story wall versus a lower story wall. The stability/capability of shear walls to withstand lateral forces may be provided at least in part by the more robust configuration of the spigots **306** used to support the shear walls. For example and as shown and described above, shear walls may be attached to spigots **306** with stronger brace configurations and a higher number of attachment (bolt) connections to tubular members of shear walls, as compared to non-shear walls that have less robust configurations for and fewer attachment connections to their respective spigots.

Furthermore, screw patterns for the metal layer **516** may vary (e.g., relatively closer versus relatively further spacing between screws) between shear walls and non-shear walls. For example, shear walls may use relatively closer screw spacing compared to non-shear walls. Progressively wider screw spacing may also be used for walls, going from lower stories to upper stories of the building.

In other embodiments, the shapes, dimensions, and configurations of studs and support members at their ends (which may not necessarily be tubular in shape in some embodiments) may vary between shear and non-shear walls, and/or may vary between floor levels. For instance, thicker stud gauges and/or more robust configurations of support members at the ends of the walls may be present for shear walls as compared to non-shear walls. Thinner and/or otherwise relatively less stronger configurations also may be present when going from a lower floor level to an upper floor level.

Such an arrangements described above may be due to the walls on upper floors having to bear less (lighter) loads than walls on lower floors, and so stud arrangements/thicknesses that can withstand smaller vertical loads may be used for upper levels of a building. Moreover, the less robust stud arrangement(s) can be used at the upper levels since there may be less shearing forces at the upper floor levels relative to lower floor levels.

The present disclosure is not to be limited in terms of the particular embodiments described in this application, which are intended as illustrations of various aspects. Many modifications and embodiments can be made without departing from its spirit and scope. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, are possible from the foregoing descriptions. Such modifications and embodiments are intended to fall within the scope of the appended claims. The present disclosure is to be limited only by the terms of the appended claims, along with the full scope of

equivalents to which such claims are entitled. This disclosure is not limited to particular methods, which can, of course, vary. The terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

With respect to the use of substantially any plural and/or singular terms herein, the terms can be translated from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

In general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.).

If a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation, no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations).

Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). Virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

In addition, where features or aspects of the disclosure are described in terms of Markush groups, the disclosure is also thereby described in terms of any individual member or subgroup of members of the Markush group.

For any and all purposes, such as in terms of providing a written description, all ranges disclosed herein also encompass any and all possible subranges and combinations of subranges thereof. Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, etc. As a non-limiting example, each range discussed herein can be readily broken down into a lower third, middle third and upper third, etc. All language such as “up to,” “at least,” “greater than,” “less than,” and the like include the number recited and refer to ranges which can be subsequently broken down into subranges as discussed above. Finally, a range includes each individual member. Thus, for example, a group having 1-3 items refers to groups having 1, 2, or 3 items. Similarly, a group having 1-5 items refers to groups having 1, 2, 3, 4, or 5 items, and so forth.

The herein described subject matter sometimes illustrates different components contained within, or connected with, different other components. Such depicted architectures are merely embodiments, and in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “operably connected”, or “operably coupled”, to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “operably couplable”, to each other to achieve the desired functionality. Specific embodiments of operably couplable include but are not limited to physically mateable and/or physically interacting components.

While various aspects and embodiments have been disclosed herein, other aspects and embodiments are possible. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting.

What is claimed is:

1. A pre-manufactured load bearing wall for a multi-story building, the load bearing wall comprising:
  - a plurality of parallel vertical metal studs;
    - a first track affixed to an upper end of the studs
    - a second track affixed to a lower end of the studs;
    - a first vertical support member at a first edge of the wall;
    - a second vertical support member at a second edge of the wall;
  - a metal plate that runs between the first and second edges of the wall, and affixed to the first and second vertical support members and to the first track, wherein the plate is configured to support a horizontal section of a first angle of a floor-ceiling panel such that the floor-ceiling panel is hung from the load bearing wall;
  - a horizontal member that runs between the first and second edges of the wall, that is affixed to the second track, and that extends laterally away from the second track, wherein the horizontal member is configured to be affixed to a second angle of the floor-ceiling panel to provide a load path for a lateral load; and
  - a stiffener device located at a top end of each of the first and second vertical support members, wherein the stiffener device is configured to provide rigidity for a joint between the first and second vertical support members and corresponding vertical support members

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respectively positioned serially above the first and second vertical support members.

2. The load bearing wall of claim 1, wherein the first and second vertical support members each comprise a hollow metal tube.

3. The load bearing wall of claim 1, wherein the first and second vertical support members are each affixed to a respective outer stud amongst the studs.

4. The load bearing wall of claim 1, wherein the first and second vertical support members each comprise a hollow metal tube, and wherein the stiffener device comprises:

a cap sized and shaped to fit an opening at the top end of the tube; and

a protrusion that extends through a slot formed in a wall of the tube at the top end of the tube.

5. The load bearing wall of claim 1, wherein the first and second vertical support members each comprise a hollow metal tube, and wherein an opening at a bottom end of the tube is sized and shaped to accommodate a spigot that is configured to secure and align the load bearing wall to an underlying structure.

6. The load bearing wall of claim 5, wherein the tube has a wall formed with holes to accommodate a corresponding plurality of fasteners to fixedly attach the tube to the spigot.

7. The load bearing wall of claim 1, wherein the studs are spaced apart by about 16 inches.

8. The load bearing wall of claim 1, further comprising a metal layer affixed to the studs, to the first and second tracks, and to outer studs respectively attached to the first and second vertical support members, wherein:

the metal layer is affixed by a first set of screws to the first and second tracks and to the outer studs, wherein the first set of screws affix the metal layer to a perimeter of the load bearing wall according to a first screw pattern, the metal layer is affixed by a second set of screws to the studs according to a second screw pattern, and the first screw pattern includes more closely spaced screws relative to the second screw pattern.

9. The load bearing wall of claim 8, wherein the metal layer is affixed by the first and second screws to only the first and second tracks, the studs, and the outer studs, and wherein the load bearing wall further comprises:

an outer layer that overlies the metal layer, wherein the outer layer is affixed by screws to only the metal layer, and is not affixed by screws to the first and second tracks, the studs, and the outer studs.

10. The load bearing wall of claim 1, wherein the plate has holes formed therein for alignment with corresponding holes formed in the horizontal section of the first angle of the floor-ceiling panel.

11. The load bearing wall of claim 1, wherein the first and second vertical support members each comprise a plurality of studs bundled together.

12. The load bearing wall of claim 1, wherein the load bearing wall is a demising wall that is not a shear wall.

13. The load bearing wall of claim 1, wherein the load bearing wall is an end wall that is also a shear wall.

14. A multi-story building, comprising:

a pre-manufactured floor-ceiling panel; and

a pre-manufactured load bearing wall that includes:

a plurality of parallel vertical metal studs;

a first track affixed to an upper end of the studs

a second track affixed to a lower end of the studs;

a first vertical support member at a first edge of the wall;

a second vertical support member at a second edge of the wall;

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a metal plate that runs between the first and second edges of the wall, and affixed to the first and second vertical support members and to the first track, wherein the plate is configured to support a horizontal section of a first angle of the floor-ceiling panel such that the floor-ceiling panel is hung from the load bearing wall;

a horizontal member that runs between the first and second edges of the wall, that is affixed to the second track, and that extends laterally away from the second track, wherein the horizontal member is configured to be affixed to a second angle of the floor-ceiling panel to provide a load path for a lateral load; and

a stiffener device located at a top end of each of the first and second vertical support members, wherein the stiffener device is configured to provide rigidity for a joint between the first and second vertical support members and corresponding vertical support members respectively positioned serially above the first and second vertical support members.

15. The building of claim 14, wherein the first and second vertical support members each comprise a hollow metal tube, and wherein the building further comprises:

a spigot configured to secure and align the load bearing wall to an underlying structure, wherein an opening at a bottom end of the tube is sized and shaped to accommodate the spigot, and wherein the tube has a wall formed with holes to accommodate a corresponding plurality of fasteners to fixedly attach the tube to the spigot.

16. The building of claim 14, wherein the building includes a plurality of the load bearing walls at each story of the building, and wherein structural strength of the load bearing walls to handle vertical and lateral loads increases from upper stories to lower stories of the building.

17. A method to manufacture a load bearing wall for a multi-story building, the method comprising:

affixing a first track to an upper end of a plurality of parallel vertical metal studs;

affixing a second track to a lower end of the studs;

affixing a first vertical support member to a first outer stud at a first edge of the wall;

affixing a second vertical support member to a second outer stud at a second edge of the wall;

affixing a metal plate, which runs between the first and second edges of the wall, to the first and second vertical support members and to the first track, wherein the plate is configured to support a horizontal section of a first angle of a floor-ceiling panel such that the floor-ceiling panel is hung from the load bearing wall;

affixing a horizontal member, which runs between the first and second edges of the wall, to the second track such that the horizontal member extends laterally away from the second track, wherein the horizontal member is configured to be affixed to a second angle of the floor-ceiling panel to provide a load path for a lateral load; and

placing a stiffener device at a top end of each of the first and second vertical support members, wherein the stiffener device is configured to provide rigidity for a joint between the first and second vertical support members and corresponding vertical support members respectively positioned serially above the first and second vertical support members.

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18. The method of claim 17, wherein the first and second vertical support members each comprise a hollow metal tube, and wherein the method further comprises:

fitting a cap into an opening at the top end of the tube;  
forming a slot in a wall of the tube at the top end of the tube; and  
extending a protrusion through the slot.

19. The method of claim 17, wherein the first and second vertical support members each comprise a hollow metal tube, wherein an opening at a bottom end of the tube is sized and shaped to accommodate a spigot that is configured to secure and align the load bearing wall to an underlying structure, and wherein the method further comprises forming holes in wall of the tube to accommodate a corresponding plurality of fasteners to fixedly attach the tube to the spigot.

20. The method of claim 17, further comprising:  
affixing a metal layer to the studs, to the first and second tracks, and to outer studs respectively attached to the first and second vertical support members, wherein:  
the metal layer is affixed by a first set of screws to the first and second tracks and to the outer studs, wherein the first set of screws affix the metal layer to a perimeter of the load bearing wall according to a first screw pattern,  
the metal layer is affixed by a second set of screws to the studs according to a second screw pattern, and  
the first screw pattern includes more closely spaced screws relative to the second screw pattern.

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